

Characteristic-based Risk Decomposition and Mispricing in High-frequency Return Panels*

Youngmin Choi[†]

Soohun Kim[‡]

Suzanne S. Lee[§]

March 2026

Abstract

We propose a novel framework that integrates firm characteristics into the estimation of systematic diffusive and jump risks using high-frequency return panels, allowing us to identify mispricing that is orthogonal to the decomposed risks. Applying the framework to more than 6,600 U.S. equities from 2000–2021, we estimate characteristic-based diffusive and jump factor betas and identify a mispricing component. We show that diffusive risk, jump risk, and mispricing are associated with distinct clusters of firm characteristics, revealing differences in their economic sources and time-series variation. The resulting intraday arbitrage portfolio delivers substantial risk-adjusted returns. We further show that the characteristic-based diffusive and jump factors command significant risk premia, with signed jump risks playing economically distinct roles.

JEL classification codes: G01, G23, G28

Keywords: Jump diffusion, characteristic-based model, projected PCA, systematic risk decomposition, mispricing

*We are grateful to Torben Andersen, Tim Bollerslev, Viktor Todorov, Roberto Reno, Per Mykland, Ruey Tsay, Xinghua Zheng, Yingying Li, Jihyun Kim, Christina Wang, Andreas Neuhierl, and seminar and conference participants at the University College Dublin, Erasmus University Rotterdam, the Midwest Finance Association (MFA), Eastern Finance Association (EFA), Society for Financial Econometrics (SoFiE), Financial Management Association (FMA), and Stevanovich Center at the University of Chicago, for their valuable suggestions, comments, and encouragement. Any remaining errors are solely our own.

[†]Williams College of Business, Xavier University. Email: youngmin.choi83@gmail.com.

[‡]College of Business, Korea Advanced Institute of Science and Technology. Email: soohun.kimi@gmail.com.

[§]Scheller College of Business, Georgia Institute of Technology. Email: suzanne.lee@scheller.gatech.edu.

1 Introduction

Explaining the cross-section of stock returns has long been a central goal of financial economics. A large body of literature shows that firm characteristics play a critical role in explaining variation in stock returns across firms. Yet understanding how firm characteristics systematically shape asset prices at high frequencies remains an important but unresolved challenge in the asset pricing literature. Recent advances demonstrate that high-frequency returns enable tracking of time-varying risk dynamics and the decomposition of systematic risk into diffusive and jump components, which carry distinct risk premia (e.g., Bollerslev et al., 2016; Aït-Sahalia et al., 2025).¹ Existing models do not integrate firm characteristics into the high-frequency risk structure or mispricing in cross-sectional settings, although studies using low-frequency returns have extensively documented their importance. This gap raises a fundamental question: how do firm characteristics separately map into higher-frequency mispricing and decomposed risk exposures, which are particularly difficult to disentangle using low-frequency returns? In this paper, we aim to address this question by proposing a framework that uses high-frequency data.

We address this inference challenge by integrating firm characteristics into the decomposition of systematic diffusive and jump risks using high-frequency return panels, and into the identification of mispricing orthogonal to the decomposed risks. Ultimately, our framework using both high-frequency data and characteristics can be applied to test the Arbitrage Pricing Theory (APT) which posits that expected returns are determined by exposures to systematic risk factors. Using high frequency data, we overcome the empirical challenge of

¹Numerous asset pricing studies have provided evidence that jump risks are priced differently from diffusive risks. For example, Bollerslev et al. (2016) and Aït-Sahalia et al. (2025) estimate the risk premia associated with continuous betas versus jump betas and report that they are priced differently. Bandi and Renò (2022) report beta in the tails, suggesting that the overall systematic risk of the hedge fund is driven by the higher systematic risk associated with return discontinuities. In addition, Pelger (2020) investigates systematic factors by principal component analysis and reports that the jump factor exhibits a pricing behavior different from that of diffusive factors. The literature broadly acknowledges the importance of market jumps for risk assessment, portfolio management, option pricing, and trading strategies, among others. See early studies such as Merton (1973), Bakshi et al. (1997), Duffie et al. (2000), and Pan (2002), Eraker et al. (2003), among others.

separately quantifying the time-varying nature of decomposed factor risk exposures as different functions of firm characteristics. This identification with different functions of characteristics for different risk types and mispricing is important for various financial applications such as pricing, investments, risk, and portfolio management, among others.

In the existing studies for the cross-section of stock returns through the lens of firm characteristics, observable characteristics are widely used as proxies for risk exposures or mispricing.² Empirical implementations typically transform characteristics into factors, such as size or value, or incorporate them directly into statistical factor models (e.g., Kozak et al., 2018; Kelly et al., 2019; Kim et al., 2021). While this literature has greatly advanced our understanding of how characteristics relate to risks and returns, most existing approaches rely on low-frequency data and therefore have limited ability to capture high-frequency risk dynamics, particularly those associated with jumps. Incorporating jump risk is important because it captures risk premiums associated with nonlinear pricing effects, which can be substantial.

The intuition behind our proposed framework is simple. Instead of using raw high-frequency returns, which may include noise in the rapidly changing real-time financial markets, we suggest first extracting more fundamentally oriented return components by filtering raw returns with characteristics before estimating latent systematic factors. The key outcome variables of this characteristics-based approach are diffusive betas, jump betas, and mispricing alphas as functions of characteristics. Our approach can be considered as an extension of the PPCA frameworks of Fan et al. (2016) and Kim et al. (2021), which rely on low-frequency data and therefore cannot distinguish between diffusive and jump risks in real-time markets. Applying our approach to our high-frequency return panels reveals unique evidence that these decomposed risks are linked to distinct sets of firm characteristics and uncovers short-lived mispricing signals that largely disappear when returns are aggregated

²Prior research highlights the staleness issue in systematic risk measures. Specifically, Daniel and Titman (1997) and Ferson and Harvey (1997) emphasize that characteristics and systematic factor loadings are collinear and that characteristics are better predictors for betas. However, existing studies do not consider how systematic diffusive and jump risk decomposition is related to characteristics in a high-frequency context.

to lower frequencies.

Specifically, our method proceeds in four steps. First, we decompose high-frequency returns into diffusive and jump components using intraday jump detection.³ Second, we project each component onto the space spanned by firm characteristics and apply PCA to the projected returns. This step filters out idiosyncratic noise and diversifiable co-jumps, which is particularly important for jump returns where observations are sparse.⁴ Third, we estimate mispricing as the characteristic-based component of returns orthogonal to the estimated diffusive and jump risk exposures. Fourth, we construct an intraday arbitrage portfolio that exploits this mispricing while controlling for the decomposed systematic risks. The framework also provides a direct mapping between firm characteristics and the estimated risk exposures and mispricing components, and allows us to examine the cross-sectional risk premia associated with the identified diffusive and jump risks.

Our framework is grounded in a continuous-time model that accommodates a broad class of multifactor structures. Diffusive betas, jump betas, and mispricing are specified as distinct functions of firm characteristics. Under this framework, we establish consistency of the estimators for decomposed diffusive and jump risk exposures, mispricing, and arbitrage portfolio returns as the cross-sectional dimension grows and the sampling frequency increases. Monte Carlo simulations show strong finite-sample performance and highlight that characteristic projection substantially improves the estimation of jump betas.

We apply our methodology to high-frequency returns for more than 6,600 US individual stocks from January 2000 to December 2021, comprising 96 million intraday return observations and 153 firm characteristics from Jensen et al. (2023). Our empirical analysis yields

³Pelger (2019) leverages a similar decomposition of diffusive and jump returns to develop an inferential theory on systematic jump factors. We use intraday jump tests of Lee and Mykland (2008) for return decomposition. Our estimation results are not affected by the application of other similarly designed jump tests, such as those used in Pelger (2019), Bollerslev et al. (2016) and Aït-Sahalia et al. (2025), which can detect intraday jumps, as long as the detection power of jump tests increases to 100%, as we increase the sampling frequency of observations so that we can precisely identify jump arrival times and sizes up to intraday levels.

⁴We assume that cross-sectionally diversifiable jumps may arise when systematic jumps occur, which complicates the inference of systematic jump risks that should be fundamentally priced. Li et al. (2019) evaluate the significance of diversifiable risks in the presence of systematic jumps.

four main findings.

First, systematic diffusive and jump risks display fundamentally different dynamics. This distinction becomes visible because our framework combines high-frequency data with a large cross-section of assets. Diffusive risk is pervasive across stocks and remains relatively stable over time. Jump risk, by contrast, is episodic. In normal periods, most jumps are idiosyncratic, but during major market disruptions—such as the global financial crisis and the COVID-19 pandemic—systematic jump comovement rises sharply.

Second, our framework relates the decomposed diffusive and jump risk exposures and mispricing to firm characteristics, allowing us to trace each component to its economic sources. Aggregating individual characteristics into the 13 clusters defined by Jensen et al. (2023), we find that systematic diffusive risk is predominantly attributed to the *Low Risk* cluster—capturing market betas, volatility proxies, and trading-activity measures—and this attribution structure is both concentrated and stable over time. Systematic jump risk draws on a broader and more time-varying set of clusters: *Value*, *Investment*, and *Quality* emerge as important sources alongside *Low Risk*, and the leading clusters rotate across periods. Mispricing exhibits the most dispersed structure: no single cluster persistently dominates, and several fundamentals-related clusters—including *Value*, *Investment*, *Quality*, and *Profitability*—contribute comparable shares with pronounced time variation. This progression—from concentrated and stable diffusive risk, to broader and more dynamic jump risk, to increasingly dispersed mispricing—highlights the distinct economic forces shaping each component.

Third, we convert the estimated mispricing signals into an implementable trading strategy by constructing an intraday arbitrage portfolio. The portfolio delivers an annualized Sharpe ratio of 2.49, with monthly alphas ranging from 2.75% to 2.92% and associated *t*-statistics consistently exceeding 10, after controlling for the Fama and French (2015) five factors, the Carhart (1997) momentum factor, and the mispricing portfolio of Kim et al. (2021). It demonstrates resilience during periods of severe market stress, returning 87.7% during the global financial crisis (October 2007 to March 2009) and 5.95% during the onset of the

COVID-19 pandemic (February to March 2020).

Fourth, we examine the cross-sectional asset-pricing implications of the estimated risk factors. Using the extracted factors from a large panel of high-frequency data, we apply the continuous-time Fama–MacBeth procedure of Aït-Sahalia et al. (2025) to estimate separate risk premia for diffusive and jump components. Diffusive factors anchored to market, size, and value benchmarks earn significant positive premia. Of the two jump factors, the first—associated with market jumps—earns a significant premium for exposure to negative jumps, while exposure to positive jumps carries little premium. The second factor shows a strong asymmetry: beta to negative jumps commands a positive premium, whereas beta to positive jumps carries a negative premium, consistent with investor demand for lottery-like payoffs (e.g., Barberis and Huang, 2008). Tracing these premia to their cluster-level sources, we find that diffusive premia concentrate in classical risk attributes, whereas jump risk premia are dispersed across multiple fundamentals-related clusters, indicating that compensation for tail risk reflects firm characteristics beyond traditional risk measures.

Our study contributes to three strands of the literature. First, we extend characteristic-based factor estimation to a continuous-time setting with high-frequency return panels decomposed into jump and diffusive components. Existing projected PCA (PPCA) frameworks (Fan et al., 2016; Kelly et al., 2019; Kim et al., 2021) incorporate firm characteristics into factor estimation but operate in discrete time and therefore cannot separate jump and diffusive sources of systematic risk.⁵ Applying PPCA separately to diffusive and jump return panels extracted from intraday data reveals that the two types of risk differ markedly in their characteristic composition and dynamics. These differences only become visible once high-frequency data and explicit risk decomposition are combined.⁶

⁵There are many important contributions in the broader PCA literature. Chamberlain and Rothschild (1983) introduce an approximate factor model, and Connor and Korajczyk (1993) study the use of principal component analysis. For high-frequency data, Aït-Sahalia and Xiu (2019) extend PCA to continuous-time models, while Chen, Mykland, and Zhang (2020) develop a robust PCA framework addressing microstructure noise and asynchronous sampling. Addressing these ultra-high-frequency inference challenges is beyond the scope of this paper; instead, we employ a similar approximation based on realized quantities constructed from high-frequency data.

⁶Kim et al. (2021) apply time-series demeaning to separately identify mispricing and beta in low-frequency

Second, we contribute to the literature on high-frequency diffusive and jump risk decomposition by incorporating firm characteristics into the estimation of decomposed factor loadings and mispricing. Pelger (2020) apply PCA to decomposed high-frequency diffusive and jump returns and document distinct pricing behavior for jump factors. Bollerslev et al. (2016) and Aït-Sahalia et al. (2025) estimate separate risk premia for diffusive and jump betas. Jacod et al. (2022) document systematic jump risk not spanned by observable factors, and Yu et al. (2024) study the low-dimensional factor structure of systematic jumps. While these studies decompose risks into diffusive and jump components using high-frequency data, they do not incorporate firm characteristics into the estimation of risk exposures. Incorporating characteristics sharpens the identification of decomposed risk exposures, with the gains most pronounced for jump risk, where sparsity and diversifiable co-jumps make conventional estimators particularly fragile. The resulting exposures are economically meaningful: both diffusive and jump betas command significant risk premia, with a pronounced asymmetry in the pricing of jump risk that can be traced to specific clusters of firm characteristics.

Third, our paper contributes to the literature on betas versus characteristics (Daniel and Titman, 1997; Chordia et al., 2017). To our knowledge, this study is the first to integrate firm characteristics into the separate estimation of diffusive and jump factor loadings. Doing so improves our understanding of decomposed systematic risks, mispricing, and cross-sectional return prediction more broadly. High-frequency data allow us to capture more precisely the time-varying relationship between characteristics and each type of risk. Our cluster-level attribution results show that this decomposition is empirically consequential: diffusive risk, jump risk, and mispricing differ markedly in which characteristics drive them and in how these relationships evolve over time. A recent study by Aleti and Bollerslev (2025) uses high-frequency data to examine how fundamental news events are priced through jump

data before implementing the PPCA method of Fan et al. (2016). Our results show that a similar separation can be achieved using high-frequency returns when the diffusive and jump components are first isolated, and PPCA is applied to each component separately. As in other projected PCA approaches, our framework performs inference at the individual stock level while incorporating a large set of firm characteristics.

risk.⁷ Our approach provides a complementary perspective by mapping firm characteristics to decomposed risk exposures and mispricing within a unified estimation framework.

The remainder of this paper is organized as follows. Section 2 introduces the theoretical model framework. Section 3 provides our inference method. In Section 4, we apply our method to US stock market data and document the empirical results. Section 5 concludes.

2 Model

We set our theoretical framework to accommodate decomposed systematic diffusive and jump risk exposures as separate functions of characteristics. We denote the i -th individual stock price by P_{it} for $i = 1, \dots, N$ and $t \in [0, T]$, where T is fixed, and the instantaneous risk-free return by R_{ft} for $t \in [0, T]$. We assume that $dp_{it}(= dP_{it}/P_{it-} - R_{ft}dt)$, the instantaneous return for the i -th stock in excess of the instantaneous risk-free rate, can be represented by the following stochastic differential equation:

$$dp_{it} = \xi_i dt + \beta'_{Ci} d\mathbf{f}_{Ct} + \sigma_{it} dW_{it} + \beta'_{Ji} d\mathbf{f}_{Jt} + de_{Jit}. \quad (2.1)$$

Note that ξ_i is the instantaneous drift term, and the next two terms describe the systematic and idiosyncratic diffusive components. β_{Ci} is a $(K_C \times 1)$ vector of exposure to diffusive factors $d\mathbf{f}_{Ct} = \Sigma_{Ct}^{1/2} d\mathbf{W}_{Ct}$, where Σ_{Ct} describes the variation of diffusive factor innovations and $d\mathbf{W}_{Ct}$ is the $(K_C \times 1)$ vector of Brownian motions. $\sigma_{it} dW_{it}$ is an idiosyncratic diffusion term for the stock i , where σ_{it} is the instantaneous diffusive volatility and dW_{it} is a scalar Brownian motion. We allow dW_{it} to be correlated across different stocks, subject to restrictions specified in Condition 2 in the Internet Appendix.

The last two terms in Equation (2.1) characterize jump dynamics, where we assume finite activity jumps throughout our analysis. We first set the term $\beta'_{Ji} d\mathbf{f}_{Jt}$, where the

⁷Lee (2012) is also related, using fundamental information events to improve jump-intensity predictions in a semiparametric framework.

$(K_J \times 1)$ vector of β_{Ji} represents the factor loadings of stock i to the systematic jump factors $d\mathbf{f}_{Jt} = Y_{ft}dJ_t^*$ with J_t^* denoting the counting process for economy-wide jump arrivals. When an economy-wide jump occurs, stock i 's jump size is characterized by Y_{it}^* . We assume a factor structure of this systematic jump size: $Y_{it}^* = \beta'_{Ji}Y_{ft} + \varepsilon_{Jit}$, where Y_{ft} represents the cross-sectionally systematic jump size associated with the economy-wide jump arrival J_t^* , and ε_{Jit} represents a diversifiable cojump size. The last term de_{Jit} is set to characterize the idiosyncratic jump term and can be decomposed as $\varepsilon_{Jit}dJ_t^* + Y_{it}dJ_{it}$, where Y_{it} represents the size of a stock i -specific jump (does not coincide with an economy-wide jump) and J_{it} represents the counting process for stock i -specific jump arrivals.⁸

Notably, this proposed framework offers greater generality than does Pelger (2019), as our setup accommodates the possibility of idiosyncratic jumps occurring alongside systematic jumps. Li et al. (2019) attribute the transient dynamics in individual stock jump betas as a potential cause for such jump behavior. For further elaboration of our theory, we impose minimal conditions in which both idiosyncratic jumps and idiosyncratic diffusive terms vanish within a well-diversified portfolio.⁹

Next, we introduce matrix notations for ease of exposition. The $(N \times 1)$ vector $\boldsymbol{\xi}$ is given by $[\xi_1, \xi_2, \dots, \xi_N]'$. The $(N \times K_C)$ matrix $\mathbf{B}_C$ and $(N \times K_J)$ matrix $\mathbf{B}_J$ are given by $[\beta_{C1}, \beta_{C2}, \dots, \beta_{CN}]'$ and $[\beta_{J1}, \beta_{J2}, \dots, \beta_{JN}]'$, respectively. Without loss of generality, we express the vector of drift terms $\boldsymbol{\xi}$ with the matrices $\mathbf{B}_C$ and $\mathbf{B}_J$:

$$\boldsymbol{\xi} = \boldsymbol{\alpha} + \mathbf{B}_C\boldsymbol{\lambda}_C + \mathbf{B}_J\boldsymbol{\lambda}_J, \quad (2.2)$$

⁸The former $\varepsilon_{Jit}dJ_t^*$ represents jumps concurrently occurring with economy-wide jumps but that remain diversifiable through cross-sectional diversification, whereas the latter $Y_{it}dJ_{it}$ represents unsystematic, stock-specific jumps that occur independently from economy-wide jumps. We do not model multiple economy-wide jump arrival processes separately but instead consolidate them into a single process, J_t^* . Conditional on $dJ_t^* = 1$, we allow multiple systematic jump sizes to affect the cross-section of stocks. Since our primary interest lies in identifying the relationship between characteristics and loadings on systematic jumps, separating multiple systematic jump arrival processes is not necessary. Moreover, our setup—with a single economy-wide jump arrival process and multiple systematic jump sizes—encompasses the conventional framework of multiple jump arrival processes, each associated with a distinct systematic jump size.

⁹See Conditions 2 in the Internet Appendix for formal restrictions.

where $\boldsymbol{\lambda}_C$ is a $(K_C \times 1)$ vector, $\boldsymbol{\lambda}_J$ is a $(K_J \times 1)$ vector, and $\boldsymbol{\alpha}$ is an $(N \times 1)$ vector. Given that ξ_i may differ from the expected instantaneous returns for stock i , the coefficients of $\boldsymbol{\lambda}_C$ and $\boldsymbol{\lambda}_J$ in Equation (2.2) are not indicative of the risk premia for exposures to the systematic factors.¹⁰ Nevertheless, it is feasible to interpret $\boldsymbol{\alpha}$ in Equation (2.2) as capturing mispricings—a point we will formalize further later in this section. The arbitrage pricing theory of Ross (1976) and Chamberlain and Rothschild (1983) states that the $(N \times 1)$ mispricing vector $\boldsymbol{\alpha}$ is not substantial in an economy without arbitrages. However, we do not restrict the mispricing vector $\boldsymbol{\alpha}$ to be small but allow it to be associated with various characteristics.

The purpose of this study is to examine how the terms $\boldsymbol{\alpha}$, $\mathbf{B}_C$, and $\mathbf{B}_J$ relate to firm characteristics. Given our short- T setting, estimating risk premia is beyond the scope of this paper. In particular, the coefficients on the factor loadings $\boldsymbol{\lambda}_C$ and $\boldsymbol{\lambda}_J$ in the drift term of (2.2) are not identified. Consequently, we only partially identify the component of the drift associated with characteristics, as we demonstrate below.

To this end, we set each of the mispricing vector $\boldsymbol{\alpha}$, the diffusive beta matrix $\mathbf{B}_C$, and the jump beta matrix $\mathbf{B}_J$ to be functions of firm characteristics. Let $\mathbf{x}_i = [x_{i1}, x_{i2}, \dots, x_{iL}]'$ be the $(L \times 1)$ vector of the characteristics associated with stock i . Define the $(N \times L)$ matrix of $\mathbf{X}$, the i -th row of which is $\mathbf{x}_i'$. We assume the following regularity condition for the matrix $\mathbf{X}$:

$$\frac{\mathbf{X}'\mathbf{X}}{N} \rightarrow \mathbf{V}_X, \quad (2.3)$$

where $\mathbf{V}_X$ is an $(L \times L)$ positive definite symmetric matrix. Then, we impose the following

¹⁰Alternatively, we could introduce a jump compensator, adjusting the drift term to reflect the expected instantaneous returns. However, we choose to utilize the expression of Equation (2.1) to avoid unnecessary notation. For a thorough derivation of the pricing equation, please refer to Section 2 of Li et al. (2019).

structure for α , $\mathbf{B}_C$ and $\mathbf{B}_J$:

$$\alpha = \mathbf{X}\Theta_\alpha + \Gamma_\alpha, \quad (2.4)$$

$$\mathbf{B}_C = \mathbf{X}\Theta_{\beta_C} + \Gamma_{\beta_C}, \quad (2.5)$$

$$\mathbf{B}_J = \mathbf{X}\Theta_{\beta_J} + \Gamma_{\beta_J}, \quad (2.6)$$

where the $(L \times 1)$ vector Θ_α , the $(L \times K_C)$ matrix Θ_{β_C} , and the $(L \times K_J)$ matrix Θ_{β_J} relate the characteristic space of $\mathbf{X}$ to α , $\mathbf{B}_C$, and $\mathbf{B}_J$, respectively, and the $(N \times 1)$ vector Γ_α , the $(N \times K_C)$ matrix Γ_{β_C} , and the $(N \times K_J)$ matrix Γ_{β_J} are cross-sectionally orthogonal to the characteristic space of $\mathbf{X}$. Note that Γ_α , Γ_{β_C} and Γ_{β_J} represent the sources of alpha and continuous and jump betas that are not related to the characteristics $\mathbf{X}$. Hence, we do not need to have a complete set of characteristics to describe α , $\mathbf{B}_C$ and $\mathbf{B}_J$. The specifications of Equations (2.4)-(2.6) allow us to lose some information through residual terms.

We provide additional qualifications for this setup. First, note that the specification of Equation (2.4) allows the violation of arbitrage pricing theory (APT, Ross (1976)). Because APT implies that $\frac{1}{N}\alpha'\alpha \rightarrow 0$, it follows that $0 \leq \frac{\Theta'_\alpha \mathbf{X}'\mathbf{X}\Theta_\alpha}{N} \leq \frac{1}{N}\alpha'\alpha$, yielding $\Theta_\alpha = \mathbf{0}_L$ from $\Theta'_\alpha \mathbf{V}_X \Theta_\alpha = 0$. Our focus pertains to the scenario where Θ_α is nonzero, with an additional aim to identify the value of Θ_α .

Next, we assume asymptotic orthogonality of $\mathbf{X}\Theta_\alpha$ against $\mathbf{X}\Theta_{\beta_C}$ and $\mathbf{X}\Theta_{\beta_J}$.

Assumption 1. *It holds that $\Theta'_{\beta_C} \mathbf{V}_X \Theta_\alpha = \mathbf{0}_{K_C}$, and $\Theta'_{\beta_J} \mathbf{V}_X \Theta_\alpha = \mathbf{0}_{K_J}$.*

This assumption serves the purpose of identification and holds without loss of generality because we can always project $\mathbf{X}\Theta_\alpha$ on $\mathbf{X}\Theta_{\beta_C}$ or $\mathbf{X}\Theta_{\beta_J}$.¹¹ In terms of economics, we give maximal power to diffusive or jump betas to explain the cross-sectional variation in returns. In addition, we need some technical conditions listed in the Internet Appendix IA.1.1. For the remainder of this paper, we assume that the Assumption 1 as well as the technical

¹¹Similarly, Connor et al. (2012) assume cross-sectional orthogonality between alpha and beta for identification in their equation (6).

conditions are satisfied.

3 Inference Method

In this section, we describe our method to estimate the parameters that connect characteristics with decomposed diffusive and jump risk exposures and mispricing Θ_α , Θ_{β_C} , and Θ_{β_J} . We use discretely observed high-frequency returns and firm-characteristics data. Given a fixed time horizon of T , we denote $\mathcal{T}$ as the number of observations in $[0, T]$ and τ as the index for counting discrete intervals $\tau = 1, \dots, \mathcal{T}$.¹² We have $\mathcal{T} = \frac{T}{\Delta t}$, where Δt is the distance between two consecutive observations of price P_{it} . We let $r_{i\tau} = \left(\frac{P_{i\tau\Delta t} - P_{i(\tau-1)\Delta t}}{P_{i(\tau-1)\Delta t}} - R_{f(\tau-1)\Delta t}\Delta t \right)$ be the realized excess return between two discrete times $(\tau - 1)\Delta t$ and $\tau\Delta t$. We set the $(N \times L)$ matrix of firm-characteristics $\mathbf{X}$ constant over a fixed time interval $[0, T]$ for our theoretical discussion.¹³

Let $\mathbf{R}$ denote the $(N \times \mathcal{T})$ matrix with the (i, τ) -th element of $r_{i\tau}$. Note that $\mathbf{R}$ is decomposed into the continuous (diffusive) component $\mathbf{R}_C$ and the jump component $\mathbf{R}_J$:

$$\mathbf{R} = \mathbf{R}_C + \mathbf{R}_J, \quad (3.1)$$

where $\mathbf{R}_C$ can be expressed as follows:

$$\mathbf{R}_C = \boldsymbol{\xi}\Delta t\mathbf{1}'_{\mathcal{T}} + \mathbf{B}_C\mathbf{F}_C + \mathbf{E}_C, \quad (3.2)$$

where $\mathbf{1}_{\mathcal{T}}$ is the $(\mathcal{T} \times 1)$ vector of ones, $\mathbf{F}_C$ is the $(K_C \times \mathcal{T})$ matrix with the (i, τ) -th element representing $\int_{(\tau-1)\Delta t}^{\tau\Delta t} d\mathbf{f}_{Ct}$: the diffusive factor realization over the τ -th subinterval, and $\mathbf{E}_C$ is the $(N \times \mathcal{T})$ matrix with the τ -th column representing $\int_{(\tau-1)\Delta t}^{\tau\Delta t} \sigma_{it}dW_{it}$: the idiosyncratic

¹²As in many other papers that use a finite T setup (e.g., Kim and Skoulakis, 2018; Raponi et al., 2020; Kim et al., 2021; Yu et al., 2024), we do not consider the randomness in factors for our analysis.

¹³We confirm the robustness of our method to realistic dynamics in $\mathbf{X}$ through a simulation exercise.

diffusive residuals over the τ -th subinterval. $\mathbf{R}_J$ can be expressed as follows:

$$\mathbf{R}_J = \mathbf{B}_J \mathbf{F}_J + \mathbf{E}_J, \quad (3.3)$$

where $\mathbf{F}_J$ is the $(K_J \times \mathcal{T})$ matrix with the (i, τ) -th element representing $\int_{(\tau-1)\Delta t}^{\tau\Delta t} d\mathbf{f}_{Jt}$: the jump factor realization over the τ -th subinterval, and $\mathbf{E}_J$ is the $(N \times \mathcal{T})$ matrix with the τ -th column representing $\int_{(\tau-1)\Delta t}^{\tau\Delta t} de_{Jit}$: the idiosyncratic jump residuals over the τ -th subinterval.

Furthermore, we can decompose $\mathbf{R}$ into a component related to characteristics and a component orthogonal to characteristics. This decomposition will serve as a useful tool for our analyses. In particular, by plugging Equations (2.4)-(2.6) into Equations (3.1)-(3.3), we obtain the following expressions:

$$\begin{aligned} \mathbf{R} &= \mathbf{X}\Theta_\alpha \Delta t \mathbf{1}'_{\mathcal{T}} + \mathbf{X}\Theta_{\beta_C} (\mathbf{F}_C + \boldsymbol{\lambda}_C \Delta t \mathbf{1}'_{\mathcal{T}}) \\ &\quad + \mathbf{X}\Theta_{\beta_J} (\mathbf{F}_J + \boldsymbol{\lambda}_J \Delta t \mathbf{1}'_{\mathcal{T}}) + \mathbf{U}, \end{aligned} \quad (3.4)$$

where

$$\begin{aligned} \mathbf{U} &= \Gamma_\alpha \Delta t \mathbf{1}'_{\mathcal{T}} + \Gamma_{\beta_C} (\mathbf{F}_C + \boldsymbol{\lambda}_C \Delta t \mathbf{1}'_{\mathcal{T}}) + \mathbf{E}_C \\ &\quad + \Gamma_{\beta_J} (\mathbf{F}_J + \boldsymbol{\lambda}_J \Delta t \mathbf{1}'_{\mathcal{T}}) + \mathbf{E}_J. \end{aligned} \quad (3.5)$$

We achieve the goal of learning Θ_α , Θ_{β_C} , and Θ_{β_J} and constructing an intraday arbitrage portfolio in four steps. In the first step, we separate the continuous component $\mathbf{R}_C$ and the jump component $\mathbf{R}_J$ in Equation (3.1) using the method from Lee and Mykland (2008). We learn Θ_{β_C} and Θ_{β_J} in the second step by applying the PPCA method to the separated $\mathbf{R}_C$ and $\mathbf{R}_J$ from the first step, respectively. In the third step, we learn Θ_α by regressing the aggregated return on $\mathbf{X}$ with the constraint that the projection is orthogonal to the estimated $\mathbf{X}\Theta_{\beta_C}$ and $\mathbf{X}\Theta_{\beta_J}$, as justified by Assumption 1. In the last step, we hold a portfolio proportional to the estimated $\mathbf{X}\Theta_\alpha$ (the third step) and confirm that the proposed

portfolio generates arbitrage profits. Methodologically, we extend the PPCA method to estimate not only the loadings on diffusive factors but also the loadings on jump factors as well as the mispricing function, a case not covered in Fan et al. (2016) and Kim et al. (2021). Each of the four steps is detailed in the remainder of this section.

3.1 First step: Return panel decomposition into $\mathbf{R}_C$ and $\mathbf{R}_J$

For return panel decomposition, we exploit a jump detection method, which distinguishes jump arrivals by identifying relatively large high-frequency returns compared to the spot volatility estimates of diffusive terms. Our jump test hinges on local normality in the absence of jumps, a consequence of the driving Brownian motion. If jumps are present, the test rejects the null of no jumps consistently.¹⁴

Specifically, we let $r_{i\tau}$ denote the (i, τ) -th element of $\mathbf{R}$. Then, we let $\widehat{\mathbf{R}}_J$ denote the $(N \times \mathcal{T})$ identified jump return matrix whose (i, τ) -th element is

$$r_{i\tau} \times I \left(\left| \frac{r_{i\tau}}{\widehat{\sigma}_{i\tau} \sqrt{\Delta t}} \right| \in \Omega_{\mathcal{T}}(\tilde{\alpha}) \right), \quad (3.6)$$

where $I(a)$ is an indicator function that is one if a is true, $\widehat{\sigma}_{i\tau}$ is the (estimated) spot volatility for the i -th stock applied over the τ -th subinterval, and $\Omega_{\mathcal{T}}(\alpha)$ is the rejection region for the jump test with an overall error rate of $\tilde{\alpha}$. Similarly, we construct the $(N \times \mathcal{T})$ matrix $\widehat{\mathbf{R}}_C$ such that the (i, τ) -th element of $\widehat{\mathbf{R}}_C$ is

$$r_{i\tau} \times I \left(\left| \frac{r_{i\tau}}{\widehat{\sigma}_{i\tau} \sqrt{\Delta t}} \right| \notin \Omega_{\mathcal{T}}(\tilde{\alpha}) \right). \quad (3.7)$$

The following theorem establishes the consistency of the jump detection in separating the diffusive and jump components in large cross sections of high-frequency return panels.

¹⁴In this paper, we use a detection method of Lee and Mykland (2008). Other jump tests in the high-frequency jump literature can be used, and our results in this paper are expected to hold, as long as jump returns are distinguished from non-jump returns, with the detection power increasing to 100%, as we increase the frequency of observations.

Theorem 3.1. *As $\Delta t \rightarrow 0$ and $N \rightarrow \infty$, it holds that $\text{tr} \left(\frac{1}{N} (\widehat{\mathbf{R}}_C - \mathbf{R}_C) (\widehat{\mathbf{R}}_C - \mathbf{R}_C)' \right) \xrightarrow{p} 0$ and that $\text{tr} \left(\frac{1}{N} (\widehat{\mathbf{R}}_J - \mathbf{R}_J) (\widehat{\mathbf{R}}_J - \mathbf{R}_J)' \right) \xrightarrow{p} 0$.*

3.2 Second step: Estimation of Θ_{β_C} and Θ_{β_J}

In the second step of our procedure, we estimate Θ_{β_C} and Θ_{β_J} , the parameters that govern how firm characteristics translate into diffusive and jump betas. First, we extract the information related to characteristics in non-jump and jump panels $\widehat{\mathbf{R}}_C$ and $\widehat{\mathbf{R}}_J$ obtained in the first step by projecting them separately onto the characteristics space. Using these projected returns, we apply a principal component analysis. By filtering decomposed returns via projection, we achieve our goal of finding decomposed latent factors that are economically meaningful relative to those characteristics, rather than merely capturing statistical noise. The resulting eigenvectors represent the optimal combination of the characteristics that explain the most structured variation. In essence, we identify these linking parameters to distinguish the contribution of each characteristic in building systematic diffusive and jump factors that best explain the decomposed diffusive and jump returns, respectively.

Specifically, we introduce the projection matrix $\mathbf{P}$ given by

$$\mathbf{P} = \mathbf{X} (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}'. \quad (3.8)$$

Then, we propose our estimator for Θ_{β_C} using projected non-jump return panel $\mathbf{P} \widehat{\mathbf{R}}_C$:

$$\widehat{\Theta}_{\beta_C} \equiv (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \sqrt{N} \widehat{\mathbf{V}}_C, \quad (3.9)$$

where $\widehat{\mathbf{V}}_C$ is the $(N \times K_C)$ matrix whose k -th column is the eigenvector associated with the k -th largest eigenvalue of $(\mathbf{P} \widehat{\mathbf{R}}_C) (\mathbf{P} \widehat{\mathbf{R}}_C)'$, and $\widehat{\mathbf{D}}_C$ is the $(K_C \times K_C)$ diagonal matrix

collecting the corresponding eigenvalues in the eigendecomposition as below:

$$\frac{(\widehat{\mathbf{P}\mathbf{R}_C})(\widehat{\mathbf{P}\mathbf{R}_C})'}{N} = \widehat{\mathbf{V}}_C \widehat{\mathbf{D}}_C \widehat{\mathbf{V}}_C'. \quad (3.10)$$

The intuition behind our proposed estimator for Θ_{β_C} is based on the following approximation as we increase the frequency of observations, so that the effect of drift term becomes negligible:

$$\frac{(\widehat{\mathbf{P}\mathbf{R}_C})(\widehat{\mathbf{P}\mathbf{R}_C})'}{N} = \widehat{\mathbf{V}}_C \widehat{\mathbf{D}}_C \widehat{\mathbf{V}}_C' \simeq \left(\frac{\mathbf{X}\Theta_{\beta_C}}{\sqrt{N}} \right) (\mathbf{F}_C \mathbf{F}_C') \left(\frac{\mathbf{X}\Theta_{\beta_C}}{\sqrt{N}} \right)'. \quad (3.11)$$

In short, it implies that $\widehat{\mathbf{V}}_C$, consisting of the eigenvectors of $\frac{(\widehat{\mathbf{P}\mathbf{R}_C})(\widehat{\mathbf{P}\mathbf{R}_C})'}{N}$, recovers the information about $\mathbf{X}\Theta_{\beta_C}$ with $\widehat{\mathbf{V}}_C \simeq \frac{\mathbf{X}\Theta_{\beta_C}}{\sqrt{N}}$.¹⁵

We apply a similar logic in eigendecomposition to the jump component as we did for the continuous component in Equation (3.10). Hence, for the jump factor loading parameters, we propose the following estimator of Θ_{β_J} :

$$\widehat{\Theta}_{\beta_J} \equiv (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\sqrt{N}\widehat{\mathbf{V}}_J, \quad (3.12)$$

where $\widehat{\mathbf{V}}_J$ is given by the following eigendecomposition:

$$\frac{(\widehat{\mathbf{P}\mathbf{R}_J})(\widehat{\mathbf{P}\mathbf{R}_J})'}{N} = \widehat{\mathbf{V}}_J \widehat{\mathbf{D}}_J \widehat{\mathbf{V}}_J'. \quad (3.13)$$

Here, $\widehat{\mathbf{V}}_J$ is the $(N \times K_J)$ matrix, the k -th column of which is the k -th eigenvector of

¹⁵Note that

$$\widehat{\mathbf{P}\mathbf{R}_C} \simeq \mathbf{P}\mathbf{R}_C = \mathbf{P}\xi\Delta t\mathbf{1}'_{\mathcal{T}} + \mathbf{P}\mathbf{B}_C\mathbf{F}_C + \mathbf{P}\mathbf{E}_C \simeq \mathbf{P}\xi\Delta t\mathbf{1}'_{\mathcal{T}} + \mathbf{X}\Theta_{\beta_C}\mathbf{F}_C,$$

where the first equality is from Equation (3.2) and the last approximation holds because $\mathbf{P}\mathbf{B}_C \simeq \mathbf{X}\Theta_{\beta_C}$ and $\mathbf{P}\mathbf{E}_C$ becomes ignorable. Hence, $(\widehat{\mathbf{P}\mathbf{R}_C})(\widehat{\mathbf{P}\mathbf{R}_C})' \simeq (\mathbf{X}\Theta_{\beta_C}\mathbf{F}_C)(\mathbf{X}\Theta_{\beta_C}\mathbf{F}_C)' = (\mathbf{X}\Theta_{\beta_C})(\mathbf{F}_C\mathbf{F}_C')(\mathbf{X}\Theta_{\beta_C})'$. Consequently, we obtain Equation (3.10). Under the identification restrictions in Condition 3(i), $\mathbf{F}_C\mathbf{F}_C'$ converges to a diagonal matrix and $\left(\frac{\mathbf{X}\Theta_{\beta_C}}{\sqrt{N}}\right)' \left(\frac{\mathbf{X}\Theta_{\beta_C}}{\sqrt{N}}\right) = \Theta_{\beta_C}' \frac{\mathbf{X}'\mathbf{X}}{N} \Theta_{\beta_C} \rightarrow \mathbf{I}_{K_C}$, which implies the relationship used for our proposal.

$(\widehat{\mathbf{P}\mathbf{R}}_J)(\widehat{\mathbf{P}\mathbf{R}}_J)'$, corresponding to the k -th eigenvalue, located at the k -th diagonal element of the $(K_J \times K_J)$ diagonal matrix $\widehat{\mathbf{D}}_J$. By the same intuition as in the diffusive case, the matrix $\widehat{\mathbf{V}}_J$, whose columns are eigenvectors of $\frac{(\widehat{\mathbf{P}\mathbf{R}}_J)(\widehat{\mathbf{P}\mathbf{R}}_J)'}{N}$ recovers the column space of $\mathbf{X}\Theta_{\beta_J}$ so that $\widehat{\mathbf{V}}_J \simeq \frac{\mathbf{X}\Theta_{\beta_J}}{\sqrt{N}}$.¹⁶

The following theorem establishes that our proposed estimators are consistent and the proof can be found in the Internet Appendix IA.1.2.

Theorem 3.2. *As $N \rightarrow \infty$ and $\Delta t \rightarrow 0$, $\widehat{\Theta}_{\beta_C} \xrightarrow{p} \Theta_{\beta_C}$ and $\widehat{\Theta}_{\beta_J} \xrightarrow{p} \Theta_{\beta_J}$, where $\widehat{\Theta}_{\beta_C}$ and $\widehat{\Theta}_{\beta_J}$ are given by Equations (3.9) and (3.12), respectively.*

Building upon the above regarding our estimators $\mathbf{X}\widehat{\Theta}_{\beta_C}$ and $\mathbf{X}\widehat{\Theta}_{\beta_J}$ for $\mathbf{X}\Theta_{\beta_C}$ and $\mathbf{X}\Theta_{\beta_J}$, we estimate the latent diffusive and jump factors respectively by

$$\widehat{\mathbf{F}}_C \equiv \left[(\mathbf{X}\widehat{\Theta}_{\beta_C})' (\mathbf{X}\widehat{\Theta}_{\beta_C}) \right]^{-1} (\mathbf{X}\widehat{\Theta}_{\beta_C})' (\widehat{\mathbf{P}\mathbf{R}}_C), \quad (3.14)$$

and

$$\widehat{\mathbf{F}}_J \equiv \left[(\mathbf{X}\widehat{\Theta}_{\beta_J})' (\mathbf{X}\widehat{\Theta}_{\beta_J}) \right]^{-1} (\mathbf{X}\widehat{\Theta}_{\beta_J})' (\widehat{\mathbf{P}\mathbf{R}}_J). \quad (3.15)$$

It is worth emphasizing that the estimation process in our second step departs fundamentally from that of Kim et al. (2021). In their approach based on low frequency data, time-series demeaning serves to isolate beta from mispricing remaining in low-frequency data before the application of the PPCA method proposed by Fan et al. (2016). The above theorem establishes that this separation is equally attainable in the realm of high-frequency returns—once the continuous and jump components are carefully isolated, and PPCA is applied to each in turn.

¹⁶Similarly, we have

$$\widehat{\mathbf{P}\mathbf{R}}_J \simeq \mathbf{P}\mathbf{R}_J = \mathbf{P}\mathbf{B}_J\mathbf{F}_J + \mathbf{P}\mathbf{E}_J \simeq \mathbf{X}\Theta_{\beta_J}\mathbf{F}_J,$$

where the first equality is from Equation (3.3) and the last approximation is valid because $\mathbf{P}\mathbf{B}_J \simeq \mathbf{X}\Theta_{\beta_J}$ and $\mathbf{P}\mathbf{E}_J$ becomes small.

3.3 Third step: Estimation of Θ_α

In this subsection, we proceed to estimate Θ_α , which links firm characteristics to mispricing. First, we aggregate the return of $\mathbf{R}$ by postmultiplying by the $(\mathcal{T} \times 1)$ vector $\mathbf{1}_\mathcal{T}$.¹⁷ From Equation (3.4), the $(N \times 1)$ vector of aggregate returns can be expressed as follows:

$$\mathbf{R}\mathbf{1}_\mathcal{T} = \mathbf{X}\Theta_\alpha T + \mathbf{X}\Theta_{\beta_C} (\mathbf{F}_C \mathbf{1}_\mathcal{T} + \boldsymbol{\lambda}_C T) + \mathbf{X}\Theta_{\beta_J} (\mathbf{F}_J \mathbf{1}_\mathcal{T} + \boldsymbol{\lambda}_J T) + \mathbf{U}\mathbf{1}_\mathcal{T}, \quad (3.16)$$

which gives

$$\frac{\mathbf{R}\mathbf{1}_\mathcal{T}}{T} = \mathbf{X}\Theta_\alpha + \mathbf{X}\Theta_{\beta_C} \left(\frac{\mathbf{F}_C \mathbf{1}_\mathcal{T}}{T} + \boldsymbol{\lambda}_C \right) + \mathbf{X}\Theta_{\beta_J} \left(\frac{\mathbf{F}_J \mathbf{1}_\mathcal{T}}{T} + \boldsymbol{\lambda}_J \right) + \frac{\mathbf{U}\mathbf{1}_\mathcal{T}}{T}. \quad (3.17)$$

Our objective is to extract Θ_α in the first term of RHS in Equation (3.17). The expression of Equation (3.5) shows that $\frac{\mathbf{U}\mathbf{1}_\mathcal{T}}{T}$, the last term of RHS in Equation (3.17), is orthogonal to $\mathbf{X}$. However, simply projecting $\frac{\mathbf{R}\mathbf{1}_\mathcal{T}}{T}$ to the linear span of $\mathbf{X}$ will not yield the desired result because $\frac{\mathbf{R}\mathbf{1}_\mathcal{T}}{T}$ contains other terms involving $\mathbf{X}\Theta_{\beta_C}$ and $\mathbf{X}\Theta_{\beta_J}$. That is, projecting $\frac{\mathbf{R}\mathbf{1}_\mathcal{T}}{T}$ to the linear span of $\mathbf{X}$ confounds the cross-sectional predictability of returns due to mispricing with the predictability of returns due to diffusive and jump factor risk premia. Hence, we project $\frac{\mathbf{R}\mathbf{1}_\mathcal{T}}{T}$ to the linear space spanned by $\mathbf{X}$ being orthogonal to $\mathbf{X}\hat{\Theta}_{\beta_C}$ and $\mathbf{X}\hat{\Theta}_{\beta_J}$ with a proper time scaling.

We propose our estimator $\hat{\Theta}_\alpha$ for Θ_α as the solution of the following constrained optimization problem:

$$\begin{aligned} \hat{\Theta}_\alpha &\equiv \arg \min_{\Theta_\alpha} \left(\frac{\mathbf{R}\mathbf{1}_\mathcal{T}}{T} - \mathbf{X}\Theta_\alpha \right)' \left(\frac{\mathbf{R}\mathbf{1}_\mathcal{T}}{T} - \mathbf{X}\Theta_\alpha \right) \\ \text{subject to } & \left[\mathbf{X}\hat{\Theta}_{\beta_C} \quad \mathbf{X}\hat{\Theta}_{\beta_J} \right]' \mathbf{X}\Theta_\alpha = \mathbf{0}_{(K_C+K_J)}, \end{aligned} \quad (3.18)$$

where $\hat{\Theta}_{\beta_C}$ and $\hat{\Theta}_{\beta_J}$ are given by the estimators in Theorem 3.2. This minimization problem is a conventional ordinary least square problem with linear equality constraints and a closed-

¹⁷We can weight the time series mean by postmultiplying any $(\mathcal{T} \times 1)$ vector $\mathbf{i}$ such that $\mathbf{1}'_\mathcal{T} \mathbf{i} = \mathcal{T}$.

form solution is easily obtained. Finally, the following theorem establishes that we can consistently estimate Θ_α with the suggested approach, and its proof can be found in the Internet Appendix IA.1.2.

Theorem 3.3. *As $N \rightarrow \infty$ and $\Delta t \rightarrow 0$, $\hat{\Theta}_\alpha \xrightarrow{p} \Theta_\alpha$, where $\hat{\Theta}_\alpha$ is given by Equation (3.18).*

3.4 Fourth step: Construction of the intraday arbitrage portfolio

In the previous subsections, we show that our approach allows consistent estimation of factor loadings on both diffusive and jump risk, as well as mispricing as a function of characteristics. In this subsection, we suggest how to exploit our characteristic-based mispricing information optimally to construct an intraday arbitrage portfolio.

Specifically, given that $\hat{\Theta}_\alpha$ summarizes the relationship between characteristics and mispricing, we propose implementing a feasible portfolio whose weight vector is defined as $\hat{\mathbf{w}} = \frac{1}{N} \mathbf{X} \hat{\Theta}_\alpha$, where $\hat{\Theta}_\alpha$ is estimated as suggested in the third step. If Θ_α is nonzero, indicating the existence of arbitrage in the cross-section of markets, our proposed intraday arbitrage portfolio $\hat{\mathbf{w}}$ is expected to collect and approximately capture a sure profit over the fixed investment horizon T . Importantly, our proposed intraday arbitrage portfolio simultaneously aggregates mispricing information available in the cross-section after considering all the corresponding diffusive and jump risks that can be systematically priced and explained by firm characteristics.¹⁸

¹⁸To explain the mechanism behind our proposal, consider first the true but unknown (and thus infeasible) intraday arbitrage portfolio, $\mathbf{w} = \frac{1}{N} \mathbf{X} \Theta_\alpha$, where N is the number of assets to be included in the portfolio and Θ_α is the true parameter for mispricing. Then, from Equation (3.16), we find that the return of this infeasible portfolio over the horizon T is given by

$$\begin{aligned} \mathbf{w}' \mathbf{R} \mathbf{1}_T &= \frac{1}{N} \Theta'_\alpha \mathbf{X}' \mathbf{X} \Theta_\alpha T + \frac{1}{N} \Theta'_\alpha \mathbf{X}' \mathbf{X} \Theta_{\beta_C} (\mathbf{F}_C \mathbf{1}_T + \boldsymbol{\lambda}_C T) \\ &\quad + \frac{1}{N} \Theta'_\alpha \mathbf{X}' \mathbf{X} \Theta_{\beta_J} (\mathbf{F}_J \mathbf{1}_T + \boldsymbol{\lambda}_J T) + \frac{1}{N} \Theta'_\alpha \mathbf{X}' \mathbf{U} \mathbf{1}_T. \end{aligned} \quad (3.19)$$

From Assumption 1, the second and third terms in the RHS of Equation (3.19) converge to zero. Furthermore, from the expression of Equation (3.5), $\mathbf{U}$ should be orthogonal to $\mathbf{X}$, implying that the last term also disappears under reasonable regularity conditions. Hence, $\mathbf{w}' \mathbf{R} \mathbf{1}_T \xrightarrow{p} \Theta'_\alpha \mathbf{V}_X \Theta_\alpha T$, where $\frac{\mathbf{X}' \mathbf{X}}{N} \rightarrow \mathbf{V}_X$, as N becomes large. This property also holds for our proposed arbitrage portfolio returns.

The following theorem shows the asymptotic property of our proposed intraday arbitrage portfolio returns, and its proof can be found in the Internet Appendix IA.1.2.

Theorem 3.4. *Define the intraday arbitrage portfolio weights to be $\hat{\mathbf{w}} = \frac{1}{N}\mathbf{X}\hat{\Theta}_\alpha$, where $\hat{\Theta}_\alpha$ is given in Theorem 3.3. As $N \rightarrow \infty$ and $\Delta t \rightarrow 0$, $\hat{\mathbf{w}}'\mathbf{R}\mathbf{1}_T/T \xrightarrow{p} \delta \geq 0$, where $\delta = \Theta'_\alpha \mathbf{V}_X \Theta_\alpha$, and $\frac{\mathbf{X}'\mathbf{X}}{N} \rightarrow \mathbf{V}_X$ and T is the investment time horizon.*

This theorem states that an investor can consistently recover arbitrage profits over an investment horizon of T . In particular, this theorem is useful in practice because our estimator does not require a large T . Hence, we can estimate $\mathbf{w}$ over one sample and calculate out-of-sample returns over a subsequent sample for practical application. The details of our empirical out-of-sample application are described in Section 4.

Remark. To assess the finite-sample performance of the estimators, we conduct a Monte Carlo exercise using simulated high-frequency return panels. The results show that incorporating firm characteristics improves estimation accuracy and sharpens identification of the underlying factor structure. Detailed simulation design and results are reported in the Internet Appendix IA.2.

4 Empirical analysis

This section examines the systematic factor structure of intraday return panels from the US equity market. Building on the framework developed in the previous section, we construct an intraday arbitrage portfolio and evaluate its performance. Finally, we assess the risk premia associated with the estimated risk factors and discuss their asset pricing implications.

4.1 Data on high-frequency returns and firm characteristics

Our empirical analysis employs a comprehensive dataset integrating high-frequency stock returns with firm characteristics. High-frequency returns are sourced from the Trade and

Quote (TAQ) database, covering January 2000 to December 2021.¹⁹ The primary analyses use high-frequency returns sampled at 30-minute intervals, with robustness checks at 5-minute, 15-minute, and 60-minute intervals. Overnight returns are excluded, as our estimation method relies on intraday high-frequency data. Lower-frequency data, including daily and monthly returns and trading volume, are sourced from the Center for Research in Security Prices (CRSP). To mitigate concerns related to liquidity, infrequent trading, and microstructure noise, we exclude stocks with prices below \$5 and those below the 25th percentile of monthly trading volume.²⁰

Firm characteristics are obtained from the dataset created by Jensen et al. (2023), encompassing 153 firm-level attributes, such as valuation, momentum, liquidity, profitability, and volatility.²¹ We lag all characteristic series by 1 month to ensure that the characteristic information is available at the beginning of each month. Our main dataset is constructed by merging firms present in the TAQ, CRSP, and firm characteristic datasets each month.

Table 1 provides summary statistics for the main sample. Panel A summarizes the sample coverage and the scale of the high-frequency data. The sample spans 252 months from January 2000 to December 2021 and includes an average of 1,336 firms per month, yielding 352,595 stock-month observations and over 96 million intraday return observations. Panel B reports monthly liquidity and size characteristics of the sample firms. Consistent with the applied liquidity filters, our sample contains actively traded stocks with substantial market capitalization and trading volume.

[Table 1 about here]

Panel C reports the decomposition of realized return variation into diffusive and jump

¹⁹We use the monthly TAQ product (second database) from January 2000 to December 2014 and the daily TAQ product (millisecond database) thereafter. To ensure data quality, we include only trades with TAQ correction indicators of 00 or 01, following standard data-quality filters. See the Daily TAQ Client Specifications: https://wrds-www.wharton.upenn.edu/documents/1967/Daily_TAQ_Client_Spec_v4.0.pdf.

²⁰For robustness checks, we also exclude firms below the 50th and 75th percentiles of monthly trading volume. The results, reported in the Internet Appendix IA.6, confirm the robustness of findings.

²¹Detailed descriptions are provided in Table IA.13 of the Internet Appendix. More details about data construction are in the Internet Appendix of Jensen et al. (2023). The results remain robust when using characteristics from Kim et al. (2021); these results are available upon request.

components. Realized variation (RV) is computed as the sum of squared intraday returns within each month and decomposed into continuous variation (CV) and jump variation (JV) using the jump detection procedure of Lee and Mykland (2008), where jumps are identified at the intraday level and the resulting components are aggregated to the monthly horizon. Although jumps are detected infrequently, they account for a nontrivial share of total return variation, underscoring the economic importance of jump risk in high-frequency equity returns.

4.2 Risk decomposition and firm characteristic attribution

4.2.1 Systematic risk decomposition

We apply the inference procedure developed in Section 3 to high-frequency equity excess returns. Our baseline analysis uses a monthly panel of intraday returns. For a given month, let $\mathbf{R}$ denote the $(N \times \mathcal{T})$ return matrix, where N is the number of traded stocks and $\mathcal{T}$ is the number of 30-minute intervals in that month. Following the procedure in Section 3.1, we first separate diffusive and jump components using the jump-detection method of Lee and Mykland (2008).²² This yields $\widehat{\mathbf{R}}_C$ and $\widehat{\mathbf{R}}_J$ of the same dimension as $\mathbf{R}$. We determine the number of systematic diffusive factors (K_C) and jump factors (K_J) by using the perturbed eigenvalue ratio (PER) test of Pelger (2019, 2020), which is designed for large cross-sections of high-frequency panel data. The PER test points to four diffusive factors ($K_C = 4$) and two jump factors ($K_J = 2$), which we adopt as our baseline specification.²³

Applying the procedures in Section 3.2, we obtain $\widehat{\Theta}_{\beta_C}$ and $\widehat{\Theta}_{\beta_J}$, which map firm characteristics to diffusive and jump betas, respectively. A key advantage of our large cross-sectional

²²We adopt a conservative threshold based on the 5% tail of the Gumbel distribution. To mitigate concerns on spurious jump detection due to the intraday volatility pattern documented by Andersen and Bollerslev (1997), we implement a time-of-day volatility adjustment following Lee and Wang (2020). Re-estimating key results using the adjusted procedure yields qualitatively unchanged conclusions. Implementation details and robustness results are reported in the Internet Appendix IA.6.7.

²³The Internet Appendix IA.3.1 details the implementation and explores alternative PER specifications. We confirm that the main results are robust to different choices of K_C and K_J , and these results are provided in the Internet Appendix IA.7 and IA.6.

framework is that it allows us to characterize how diffusive and jump risks separately vary across individual assets. To summarize these cross-sectional patterns, we construct a firm-level pseudo- R^2 , defined as the ratio of systematic variation to total variation in projected intraday returns—the high-frequency analogue of a time-series R^2 , computed without an intercept.²⁴

Figure 1 plots the pooled cross-sectional distributions of the pseudo- R^2 for the projected diffusive and jump returns. Panel A shows that pseudo- R^2 s for diffusive returns are concentrated at relatively high levels, with a pronounced mode around 0.7–0.8. This pattern accords with the economic interpretation of diffusive risk as reflecting pervasive common shocks and continuous comovement in equity returns. Conditional on firm characteristics, a large share of diffusive realized variation is absorbed by the systematic diffusive factor space.

[Figure 1 about here]

Panel B reveals a sharply contrasting cross-sectional pattern for the jump component. The pseudo- R^2 for the jump component is highly right-skewed, with substantial mass close to zero but a long right tail. Note that, in the absence of systematic jumps, the pseudo- R^2 would be concentrated near zero across firm-months. Instead, the right tail shows that there are periods and sets of firms for which jump returns move together strongly. This pattern suggests that the estimated systematic jump factor space captures genuine episodes of market-wide or sector-wide jump comovement rather than noise.

Figure 2 complements these pooled cross-sectional distributions by illustrating the time-series dynamics of the cross-sectional distribution of pseudo- R^2 . Panel A shows that pseudo- R^2 remains consistently high throughout the sample period, with the median pseudo- R^2 typically well above one-half and the interquartile range relatively stable. This time variation reinforces the interpretation of the diffusive factor space as capturing pervasive comovements.

²⁴For stock i , the total variations of projected diffusive and jump returns are the inner product of the i -th row of $\mathbf{P}\hat{\mathbf{R}}_C$ and $\mathbf{P}\hat{\mathbf{R}}_J$, respectively. Their systematic variations are the inner products of the i -th row of $\mathbf{X}\hat{\Theta}_{\beta_C}\hat{\mathbf{F}}_C$ and $\mathbf{X}\hat{\Theta}_{\beta_J}\hat{\mathbf{F}}_J$, where $\hat{\mathbf{F}}_C$ and $\hat{\mathbf{F}}_J$ are obtained from the cross-sectional regressions of $\mathbf{P}\hat{\mathbf{R}}_C$ and $\mathbf{P}\hat{\mathbf{R}}_J$ on $\mathbf{X}\hat{\Theta}_{\beta_C}$ and $\mathbf{X}\hat{\Theta}_{\beta_J}$ at each subinterval, respectively.

[Figure 2 about here]

Panel B shows that the pseudo- R^2 for the jump component is considerably more state dependent. In typical months, the median pseudo- R^2 remains low, and the lower tail lies close to zero, reflecting the predominance of idiosyncratic jump events. However, both the median and the upper tail spike sharply in stress episodes, and the cross-sectional dispersion widens substantially in these periods. This pattern is consistent with systematic jump risk being episodic. Taken together, Figures 1 and 2 provide a clear diagnostic that our PPCA-based decomposition produces empirically well-behaved and economically interpretable systematic diffusive and jump risk components.

4.2.2 Attribution of systematic risk to firm characteristics

Our model and inference method allow both systematic risk exposures and mispricing to be functions of firm characteristics, summarized by parameters of $\hat{\Theta}_{\beta_C}$, $\hat{\Theta}_{\beta_J}$, or $\hat{\Theta}_\alpha$. Having shown that the estimated factor spaces capture meaningful systematic variation, we now exploit our framework to attribute the estimated systematic diffusive and jump risks to firm characteristics. In this subsection, we examine how the cross-sectional variation in the estimated systematic risk components can be mapped back to observable characteristics, thereby providing an economic interpretation of the estimated diffusive and jump risk and clarifying which firm attributes are most strongly associated with each type of systematic risk.²⁵

Generic characteristic-level attribution metric. To formalize the analysis, consider a generic $(L \times M)$ matrix Ψ that maps the L -dimensional vector of firm characteristics into

²⁵We do not attempt to preclassify characteristics as exclusively diffusive-, jump-, or mispricing-related. Although we assume that $\mathbf{X}\hat{\Theta}_\alpha$ is orthogonal to $\mathbf{X}\hat{\Theta}_{\beta_C}$ and $\mathbf{X}\hat{\Theta}_{\beta_J}$, each of these factor loadings and the mispricing term can be driven by the same set of characteristics. For example, consider a single-factor return generating process for firm i : $r_i = \alpha_i + \beta_i f_t + \varepsilon_i$. Suppose that α_i and β_i depend linearly on two independent characteristics, $c_{1i} \sim N(0, 1)$ and $c_{2i} \sim N(0, 1)$, such that $\alpha_i = c_{1i} + c_{2i}$ and $\beta_i = c_{1i} - c_{2i}$. In this scenario, although α_i and β_i are orthogonal by construction, both characteristics c_{1i} and c_{2i} contribute to each component.

an M -dimensional object of interest.²⁶ Given the $(N \times L)$ matrix of firm characteristics $\mathbf{X}$, the product $\mathbf{X}\Psi$ represents the characteristic-implied object across firms. Different choices of Ψ correspond to different objects in our framework as detailed in the subsequent sections.

To quantify characteristic-level attribution, we assess how each characteristic maps into cross-sectional variation in $\mathbf{X}\Psi$. As a benchmark, we measure the overall variation of $\mathbf{X}\Psi$ using the Frobenius norm,

$$\|\mathbf{X}\Psi\| = \sqrt{\text{tr}(\mathbf{X}\Psi\Psi'\mathbf{X}')}.$$

We define the attribution of the l -th characteristic as

$$\frac{\text{tr}(\mathbf{X}i_l\Psi\Psi'\mathbf{X}')}{\|\mathbf{X}\Psi\|}, \quad (4.1)$$

where i_l is an $(L \times L)$ indicator matrix with a one in position (l, l) and zeros elsewhere. This measure captures how variation in the l -th characteristic across firms is attributed to variation in $\mathbf{X}\Psi$.

By construction, characteristic-level attributions aggregate additively:

$$\sum_{l=1}^L \frac{\text{tr}(\mathbf{X}i_l\Psi\Psi'\mathbf{X}')}{\|\mathbf{X}\Psi\|} = \|\mathbf{X}\Psi\|.$$

This property establishes a direct link between firm characteristics and their economic relevance for shaping systematic risks and mispricing.²⁷ In the empirical analysis that follows, we work with the absolute magnitudes of characteristic attributions and normalize them within each month so that they sum to one across characteristics. This normalization allows us to interpret each characteristic's attribution as its relative importance in explaining cross-sectional variation in the corresponding component at a given point in time.

²⁶ $M(< N)$ is a positive integer that depends on the application.

²⁷This result follows directly from the additive structure of the Frobenius norm.

From characteristic-level to cluster-level attribution While characteristic-level attribution provides granular information, interpretation becomes challenging given the large number of characteristics, which exceeds one hundred in our setting. We therefore aggregate individual characteristic attributions into economically meaningful clusters. Specifically, we adopt the taxonomy of Jensen et al. (2023), who construct factor returns for 153 firm characteristics and group these factors into 13 clusters exhibiting strong economic and statistical similarity.²⁸ The 13 clusters include *Accruals*, *Debt Issuance*, *Investment*, *Low Leverage*, *Low Risk*, *Momentum*, *Profit Growth*, *Profitability*, *Quality*, *Seasonality*, *Short-Term Reversal*, *Size*, and *Value*.

We utilize the clusters as follows. At each month-end, cluster attributions are computed as the sum of normalized (absolute) characteristic attributions within each cluster, yielding nonnegative cluster shares that sum to one by construction. We report the cluster-level results to facilitate economic interpretation and to highlight the time-varying dynamics of diffusive risk, jump risk, and mispricing. Detailed characteristic-level results are relegated to the Internet Appendix IA.4, with the full characteristic-cluster mapping provided in Table IA.13 in the Internet Appendix.

4.2.3 Cluster-level attribution of systematic diffusive risk

We now turn to attributing systematic diffusive risk to characteristic clusters. Specifically, we set $\Psi = \hat{\Theta}_{\beta_C} \hat{\mathbf{D}}_C^{0.5}$, where $\hat{\Theta}_{\beta_C}$ is estimated as in Section 3.2, and $\hat{\mathbf{D}}_C$ is a diagonal matrix consisting of the first $K_C = 4$ eigenvalues of $(\mathbf{P}\hat{\mathbf{R}}_C)(\mathbf{P}\hat{\mathbf{R}}_C)'$ of projected diffusive returns. Panel A of Figure 3 displays a heat map of the relative cluster-level attributions of systematic diffusive risk over the sample period. Darker shading indicates greater relative attribution among clusters at each point in time, and clusters are ordered by their average attribution

²⁸Jensen et al. (2023) first construct factor returns for each characteristic using characteristic-sorted portfolios and compute CAPM-residual factor returns. They then define the distance between factors as one minus the pairwise correlations of their CAPM-residual returns and apply hierarchical agglomerative clustering of Murtagh and Legendre (2014) using Ward (1963) linkage criterion. Based on this procedure, they select 13 clusters that exhibit a high degree of economic and statistical similarity. See their Internet Appendix for details.

of diffusive risk over the sample period. Two main patterns emerge.

[Figure 3 about here]

First, systematic diffusive risk is predominantly attributed to the *Low Risk* cluster throughout the sample period. Here, the term *Low Risk* refers to a cluster of characteristics capturing various dimensions of traditional risk exposure – such as market beta or volatility – rather than firm-specific characteristics identifying low-risk firms.²⁹ This cluster consistently accounts for the largest share of diffusive risk attribution, and its dominance is both large in magnitude and persistent over time.

Second, the cluster structure of diffusive risk is parsimonious and relatively stable. Beyond the dominant *Low Risk* cluster, other clusters—such as *Value*, *Quality*, and *Momentum*—play secondary roles and exhibit limited time variation in their attributions. In contrast to the jump and mispricing components discussed below, systematic diffusive risk displays a compact and stable attribution structure. Overall, our finding suggests that cross-sectional variation in systematic diffusive risk exposure is primarily associated with classical risk attributes.

4.2.4 Cluster-level attribution of systematic jump risk

We next examine the attribution of systematic jump risk to each cluster. To this end, we use $\Psi = \hat{\Theta}_{\beta_J} \hat{\mathbf{D}}_J^{0.5}$, where $\hat{\Theta}_{\beta_J}$ is the estimated jump loading described in Section 4.2, and $\hat{\mathbf{D}}_J$ is a diagonal matrix containing the first $K_J = 2$ eigenvalues of $(\mathbf{P}\hat{\mathbf{R}}_J)(\mathbf{P}\hat{\mathbf{R}}_J)'$. Panel B of Figure 3 displays a time-series heat map of the relative cluster-level attribution of systematic jump risk. Clusters are ordered by their average attribution of jump risk over the sample period.

²⁹The *Low Risk* cluster in Jensen et al. (2023) includes risk-related characteristics such as market beta, Frazzini and Pedersen (2014) beta, and downside beta (e.g., `beta_60m`, `betabab_1260d`, `betadown_252d`), volatility proxies (e.g., `rvol_21d`, `ivol_capm_21d`), maximum daily returns (e.g., `rmax5_21d`), and trading-activity/liquidity variables (e.g., `turnover_126d`, `zero_trades126d`). The full characteristic-cluster mapping is provided in the Internet Appendix.

Compared to the diffusive risk, the cluster-level attribution of systematic jump risk is markedly more time-varying. While the *Low Risk* cluster remains an important source of attribution on average, its dominance is substantially reduced, and several additional clusters account for meaningful shares of jump risk attribution over time.

In particular, the *Value*, *Investment*, and *Quality* clusters emerge as key sources of attribution for systematic jump risk. In the taxonomy of Jensen et al. (2023), these clusters capture valuation metrics, investment-related ratios, and measures of profitability and balance-sheet strength. Their prominence indicates that systematic jump risk is attributed not only to classical risk measures in the *Low Risk* cluster, but also to firm fundamentals that shape sensitivity to market-wide stress episodes. Another salient feature of Panel B is the pronounced time variation in cluster attribution. Leading clusters rotate across periods, with valuation- and fundamentals-related clusters accounting for larger shares of attribution in certain periods. This contrasts with the stable and parsimonious attribution structure observed for diffusive risk and underscores the dynamic nature of systematic jump risk.

Taken together, whereas systematic diffusive risk is concentrated in classical risk attributes such as betas and volatility, systematic jump risk draws on a broader and more time-varying set of firm characteristics—particularly valuation, investment, and balance-sheet quality—consistent with tail risk being shaped by firm fundamentals that become salient during market-wide stress episodes.

4.3 Mispricing and firm characteristic attribution

4.3.1 Mispricing estimation

Having identified the systematic diffusive and jump risk components, we next examine the mispricing component of returns attributable to firm characteristics. Using the monthly high-frequency panel, we estimate the mispricing coefficient vector $\hat{\Theta}_\alpha$ following Theorem 3.3 and form the firm-level mispricing component $\hat{\alpha} \equiv \mathbf{X}\hat{\Theta}_\alpha$. The i -th element $\hat{\alpha}_i$ represents the characteristic-implied pricing error for stock i in a given month.

Next, we examine the distributional properties of $\hat{\alpha}_i$, measured in basis points per 30-minute interval under our setup. Panel A of Figure 4 shows that the pooled distribution is centered close to zero and roughly symmetric. Yet the dispersion is economically meaningful. The interquartile range spans about ± 1 bp per interval, which cumulates to roughly $\pm 2.7\%$ per month.

[Figure 4 about here]

Panel B reveals that this dispersion is far from constant. Instead, it expands and contracts with market conditions. The median of $\hat{\alpha}_i$ remains anchored near zero throughout the sample, but the spread between the 10th and 90th percentiles changes markedly over time. During episodes of severe stress—most notably the global financial crisis and the COVID-19 shock—the band roughly doubles relative to calmer periods.

4.3.2 Converting mispricing to profit: Intraday arbitrage portfolio

The empirical question we now tackle is whether the estimated mispricing component translates into economically meaningful out-of-sample profits. To investigate this question, we construct an intraday arbitrage portfolio and evaluate its performance in a fully out-of-sample setting. Using data from month m , we estimate mispricing coefficients $\hat{\Theta}_\alpha$ and intraday arbitrage portfolio weights $\hat{\mathbf{w}}$ following Theorems 3.3 and 3.4, respectively.³⁰ The resulting portfolio is then implemented during the subsequent month $m + 1$. We calculate the monthly returns of our intraday arbitrage portfolio as the sum of intraday returns over month $m + 1$.³¹ By rolling the estimation window forward each month, we ensure that our portfolio’s performance is out-of-sample and free of look-ahead bias.³²

³⁰We further scale $\hat{\mathbf{w}}$ so that the resulting portfolio has an annualized volatility of 10% in the estimation month m .

³¹We revisit the performance of our intraday arbitrage portfolio as the sum of daily or monthly returns over a holding period to examine the effect of frequent rebalancing and trading costs in the Internet Appendix IA.6.

³²Our intraday arbitrage portfolio differs distinctly from the approach of Kim et al. (2021) in that it explicitly decomposes systematic diffusive and jump risks and leverages high-frequency return panel data along with firm characteristics. These methodological differences uniquely position our portfolio to effectively capture jump risk and mispricing in high-frequency return panels. In the Internet Appendix IA.6, we provide

Our intraday arbitrage portfolio delivers very strong performance with an annual Sharpe ratio of 2.49.³³ Its strong performance is particularly pronounced compared with that of the US market portfolio. Figure 5 shows the log of the time series of portfolio value, starting with \$1 invested in our intraday arbitrage portfolio at the beginning of the sample period (January 2000), compared with that from investing \$1 in the S&P 500 index.

[Figure 5 about here]

The empirical evidence shown in Figure 5 underscores the substantial economic gain generated by our intraday arbitrage portfolio. Over the entire sample period from January 2000 to December 2021, the portfolio consistently and significantly outperforms the market benchmark, achieving substantial cumulative growth. Moreover, our portfolio exhibits remarkable resilience during periods of market turbulence. Specifically, during the global financial crisis (October 2007 to March 2009), our intraday arbitrage portfolio delivered a return of 87.7%, whereas the S&P 500 index declined by 47.7%. Similarly, during the onset of the COVID-19 pandemic (February to March 2020), our portfolio generated a 5.95% return, whereas the S&P 500 index declined by 19.9%. On average, our intraday arbitrage strategy yields an annual return of 32.66%, highlighting its robustness and effectiveness across various market conditions.

To further investigate whether the performance of our intraday arbitrage portfolio can be attributed to exposure to common systematic risk factors, we reassess our intraday portfolio performance after controlling for widely recognized systematic factors. Specifically, we regress the monthly returns of our intraday arbitrage portfolio against standard asset pricing factor models, including the CAPM, the Fama and French (1993) three-factor model (FF3), the Carhart (1997) four-factor model (FF3+UMD), the Fama and French (2015) five-factor model (FF5), and the Fama and French (2018) six-factor model (FF5+UMD).³⁴ Table 2

detailed analyses confirming that our intraday arbitrage portfolio is not subsumed by the portfolio using their approach.

³³The sensitivity analyses of our arbitrage portfolio performance to the choice of empirical parameters are provided in the Internet Appendix IA.6.

³⁴Monthly data on these standard factors are obtained from the Fama-French data library in WRDS.

summarizes the monthly alpha (i.e., intercept) and factor loadings, along with corresponding t -statistics.

[Table 2 about here]

The regression results presented in Panel A of Table 2 demonstrate that the performance of our intraday arbitrage portfolio is not affected by exposure to common risk factors. Even after controlling for these widely recognized factors, the intercepts (i.e., monthly alphas) remain significantly positive across all factor-model specifications, ranging from 2.75% (FF5+UMD) to 2.92% (CAPM), with corresponding t -statistics consistently well exceeding 10. Furthermore, the explanatory power of these factor models is also limited. The adjusted R^2 remains below 10% across all regressions, indicating that the variation in the portfolio's returns is largely orthogonal to widely recognized risk factors.

We next examine whether these profits persist across time. Figure 5 suggests that the cumulative growth of the portfolio slows in the later part of the sample, particularly after the 2008 global financial crisis. A natural interpretation is that financial markets became more efficient over time, with faster information diffusion and improved price discovery compressing arbitrage opportunities. To assess whether our portfolio retains a significant alpha in this period, we analyze the post-2008 subsample (January 2009 to December 2021) and report the results in Panel B of Table 2. Although monthly alphas and their corresponding t -statistics of the subsample are lower than those of the full sample, monthly alphas remain consistently positive and statistically significant across all factor-model specifications, ranging from 1.19% to 1.26% per month with t -statistics well exceeding 4. The persistent statistical significance of the portfolio's alpha underscores the enduring economic relevance of our methodology.

In summary, our intraday arbitrage portfolio effectively utilizes high-frequency return panel data and firm characteristics to identify and exploit cross-sectional mispricing signals. Our approach delivers robust and economically significant performance, demonstrating re-

silience across diverse market conditions, including turbulent periods and the increasingly competitive post-2008 market environment.

4.3.3 Cluster-level attribution of mispricing

Next, we seek to understand where the estimated mispricing originates. To this end, we attribute the estimated mispricing component to clusters of firm characteristics. Specifically, we utilize the cluster-level attribution, described in Section 4.2.2, by setting $\Psi = \hat{\Theta}_\alpha$, where $\hat{\Theta}_\alpha$ represents the mispricing coefficients estimated previously following Theorem 3.3. Figure 6 presents a time-series heatmap of the relative cluster-level attributions of mispricing. Clusters are ordered by their average attribution of mispricing over the sample period.

[Figure 6 about here]

Relative to the diffusive and jump risks, the cluster-level attribution of mispricing is widely dispersed. No single cluster persistently accounts for a dominant share of attribution. Instead, several clusters—including *Value*, *Investment*, *Quality*, and *Profitability*—account for comparable shares of attribution, with substantial time variation in their relative importance. In the taxonomy of Jensen et al. (2023), these clusters include valuation metrics, real-side investment activity, profitability, and balance-sheet strength. Their joint prominence indicates that mispricing is attributed to a broad set of firm fundamentals rather than to a narrow economic dimension. This pattern contrasts with the more parsimonious structure observed for systematic risk, particularly for diffusive risk in Figure 3 (Panel A), where attribution concentrates in a smaller set of classical risk attributes. Mispricing, by comparison, appears inherently high-dimensional and heterogeneous. Thus, while systematic diffusive risk is concentrated in classical risk attributes and systematic jump risk draws additionally on valuation and balance-sheet characteristics, mispricing reflects a broader combination of firm fundamentals with no single dominant driver.

4.4 Cross-sectional asset pricing implication

4.4.1 Risk premia of systematic diffusive and jump risks

So far, we have shown that our inference approach separately identifies systematic diffusive and jump risk exposures based on characteristics, and that controlling for these characteristic-based risk exposures enhances our estimation of mispricing signals with economically significant out-of-sample returns. This naturally raises the next question: are these characteristic-based risk exposures themselves priced in the cross-section? To address this question, we estimate the associated risk premia. For this purpose, we embed our analysis within the continuous-time Fama–MacBeth procedure of Aït-Sahalia et al. (2025), which allows the estimation of separate premiums associated with decomposed diffusive and jump risks.³⁵

We again start from the 30-minute diffusive and jump return panels and their projections on characteristic, as constructed in Section 4.2. Using the PPCA estimators in Equations (3.14) and (3.15), we extract diffusive and jump statistical factors separately over rolling windows. Because these statistical factors are identified only up to rotation in each window, we splice the estimated factors across time to examine cross-sectional asset-pricing implications over the full sample period. Specifically, diffusive factors are estimated using a 12-month rolling window updated monthly. Adjacent rolling windows overlap for 11 months, which allows us to identify a rotation that minimizes the distance between factors over the overlapping period while preserving their covariance structures. Appending the rotated factors from the last month of each estimation window yields a factor series over the full sample.³⁶ Jump factors are spliced analogously, using a 36-month rolling window to account for the relative sparsity of jump observations. The matrices $\hat{\mathbf{F}}_C$ and $\hat{\mathbf{F}}_J$ in Equations (3.14) and (3.15) represent factor estimates within a given window. Splicing these estimates across rolling windows yields four diffusive factors, $\mathbf{f}_C^{\text{PC1}}, \mathbf{f}_C^{\text{PC2}}, \dots, \mathbf{f}_C^{\text{PC4}}$, and two jump factors, $\mathbf{f}_J^{\text{PC1}}$

³⁵Because our framework is designed for short- T settings, it is not well suited for estimating risk premia. We therefore rely on a complementary approach designed for a large- T environment.

³⁶See the Internet Appendix IA.3.2 for details on the factor splicing procedure.

and $\mathbf{f}_j^{\text{PC2}}$.³⁷

Finally, we facilitate economic interpretation by anchoring the statistical factors to well-known asset-pricing benchmarks. For the diffusive component, we first aggregate the high-frequency factors to the daily level. We then project each of the three standard daily factors—market, size, and value—onto the linear span of our four diffusive factors.³⁸ The fitted values from these projections define three anchored diffusive factors, denoted $\mathbf{f}_C^{\text{MKT}}$, $\mathbf{f}_C^{\text{SMB}}$, and $\mathbf{f}_C^{\text{HML}}$. A fourth factor, $\mathbf{f}_C^{\text{Orth}}$, is constructed as the linear combination of diffusive factors that is orthogonal to the previously identified three factors, $\mathbf{f}_C^{\text{MKT}}$, $\mathbf{f}_C^{\text{SMB}}$, and $\mathbf{f}_C^{\text{HML}}$, while remaining within the span of the original diffusive factor space. For the jump component, we project the market factor onto the span of the two jump factors to obtain a market jump factor, $\mathbf{f}_j^{\text{MKT}}$. The remaining orthogonal component, $\mathbf{f}_j^{\text{Orth}}$, captures systematic jump variation that is independent of market-wide jump risk but still lies within the same jump factor space.³⁹ We examine the pricing implications of these anchored factors.⁴⁰

Pricing of diffusive and jump factors We next examine whether the systematic diffusive and jump factors constructed above are priced in the cross-section of expected returns. Specifically, in the first stage, we estimate diffusive and jump betas from time-series regressions of projected diffusive and jump return panels on previously estimated anchored factors.⁴¹ In the second stage, we run monthly cross-sectional regressions of next-month

³⁷We provide a detailed analysis of the dynamics of these statistical diffusive and jump factors in the Internet Appendix IA.5.2.

³⁸We investigate the effect of different factor dimensions on asset pricing implications and arbitrage portfolio performance. Our findings are robust to alternative choices of factor dimensions based on PER test and are provided in the Internet Appendix IA.7.1. Daily factor data are obtained from the Fama–French data library on WRDS.

³⁹For comparability, we scale $\mathbf{f}_C^{\text{Orth}}$ and $\mathbf{f}_j^{\text{Orth}}$ to have the same variance as $\mathbf{f}_C^{\text{MKT}}$ and $\mathbf{f}_j^{\text{MKT}}$, respectively.

⁴⁰The anchored factors are linear combinations of the statistical factors extracted via PPCA and therefore span the same factor space. As a result, their cross-sectional fit coincides with that of the underlying statistical factors. The anchoring step does not alter pricing performance. Rather, it enhances economic interpretability by linking the latent factors to familiar asset-pricing benchmarks. Accordingly, we present and discuss the pricing results based on the anchored factors in the main text, while reporting the corresponding results for the statistical factors in the Internet Appendix IA.7.2 for completeness.

⁴¹Following Aït-Sahalia et al. (2025), diffusive betas are estimated on a monthly basis, while jump betas are estimated from the full sample period to accommodate the sparse and episodic nature of systematic jumps.

excess returns on the estimated diffusive and jump betas. These monthly cross-sectional regression coefficients represent estimates of the diffusive and jump risk premia. We then aggregate these coefficients over time to obtain average risk premia, and conduct inference using the procedure proposed by Ait-Sahalia et al. (2025).

Table 3 presents the risk premia associated with systematic diffusive and jump factors obtained as described above. Column (1) reports risk premium estimates from the following specification: three diffusive factors corresponding to the market, size, and value benchmarks, one orthogonal diffusive factor, and two jump factors capturing market-wide and orthogonal jump risk. The regression results suggest that exposures to the diffusive factors corresponding to market, size, and value ($\mathbf{f}_C^{\text{MKT}}$, $\mathbf{f}_C^{\text{SMB}}$, and $\mathbf{f}_C^{\text{HML}}$) earn positive and statistically significant risk premia, consistent with existing asset pricing evidence (e.g., Fama and French, 1993, 2015). The results confirm that our extracted diffusive factor space captures economically meaningful sources of systematic risk.

[Table 3 about here]

Turning to jump risk, the exposure to market jump factor $\mathbf{f}_J^{\text{MKT}}$ commands a positive and statistically significant risk premium. This result indicates that systematic jump risk is priced over and above diffusive risks. We also find that the exposure to the orthogonal jump factor, $\mathbf{f}_J^{\text{Orth}}$, is priced at 10% significance levels, suggesting that jump variation orthogonal to market-wide jump risk carries independent compensation.

Motivated by existing evidence that positive and negative jumps have distinct economic and pricing implications (e.g., Barro, 2006; Bollerslev and Todorov, 2011; Bollerslev et al., 2015), we further decompose two jump factors ($\mathbf{f}_J^{\text{MKT}}$ and $\mathbf{f}_J^{\text{Orth}}$) into positive ($\mathbf{f}_J^{\text{MKT_Pos}}$ and $\mathbf{f}_J^{\text{Orth_Pos}}$) and negative ($\mathbf{f}_J^{\text{MKT_Neg}}$ and $\mathbf{f}_J^{\text{Orth_Neg}}$) components. Column (2) of Table 3 reports the results from the extended specification, which reveals a pronounced asymmetry in jump risk pricing.

For the market jump factor, exposure to negative market jumps $\mathbf{f}_J^{\text{MKT_Neg}}$ carries a large and highly significant risk premium, whereas exposure to positive market jumps $\mathbf{f}_J^{\text{MKT_Pos}}$

is not priced. In other words, the market jump risk premium is concentrated entirely in downside risk. This pattern indicates that investors are primarily compensated for bearing downside market jump exposure, consistent with the existing literature on jump risk premia in equity markets.

A similar asymmetry arises for the orthogonal jump factor. Exposure to the negative orthogonal jump factor $\mathbf{f}_J^{\text{Orth.Neg}}$ commands a positive and statistically significant risk premium, comparable in magnitude to that of the negative market jump factor. This finding suggests the presence of an additional source of downside jump risk that is not spanned by market-wide jumps. In contrast, exposure to the positive orthogonal jump factor $\mathbf{f}_J^{\text{Orth.Pos}}$ carries a significantly negative risk premium. This negative premium is consistent with investor demand for positively skewed, lottery-like stocks that trade at elevated valuations and earn lower average returns (e.g., Barberis and Huang, 2008). Taken together, these results indicate that jump risk premia are fundamentally asymmetric and that market-wide and orthogonal jump risks represent economically distinct sources of priced tail risk.

4.4.2 Cluster-level attribution of risk premia

Next, we trace the sources of these risk premia by applying the attribution framework developed in Section 4.2. To do so, we set Ψ equal to the corresponding column of the estimated loading matrix $\hat{\Theta}_\beta$, scaled by the estimated risk premium. With this normalization, the resulting cluster-level attributions add up exactly to the total risk premium associated with each factor. Tables 4 and 5 report the decomposition results for the diffusive and jump factors, respectively.

[Table 4 about here]

[Table 5 about here]

The attribution of diffusive risk premia echoes the pattern observed for diffusive risk exposures. The premium on the diffusive market factor is heavily concentrated in the *Low*

Risk cluster, and the premium on the diffusive size factor in its namesake cluster. The premium on the diffusive value factor is dispersed across *Low Risk*, *Size*, *Momentum*, and *Quality* rather than concentrated in the *Value* cluster. This pattern reflects the fact that the *Value* cluster narrowly captures price-scaled valuation ratios (e.g., book-to-market, earnings-to-price), whereas characteristics with value-relevant economic content—such as leverage, profitability, and asset growth measures—are distributed across multiple other clusters in the taxonomy of Jensen et al. (2023).

The jump risk premia attribution exhibits a stark difference. For the negative market jump factor, the *Low Risk* cluster remains the leading contributor but its dominance is substantially reduced relative to the diffusive case, with *Size*, *Low Leverage*, *Value*, and *Profitability* all contributing meaningful and individually significant shares. The negative orthogonal jump premium is even more dispersed: *Value*, *Low Risk*, *Quality*, *Momentum*, and *Low Leverage* share comparable attributions with no single cluster dominating. This contrast indicates that compensation for downside jump risk increasingly reflects firm fundamentals as one moves beyond the market component.

4.5 Additional empirical analysis

We conduct an extensive set of additional analyses to validate the robustness of the main findings of this paper. To conserve space, detailed results—including all tables and figures—are reported in the Internet Appendix. Here, we briefly summarize the scope of these tests and their key implications.

Estimated Factor Dynamics (Internet Appendix IA.5). Projections on characteristics strengthen the identification of factor structure in high-frequency return panels. The estimated jump factors spike around major economic events, including the global financial crisis (October 2008) and the onset of the COVID-19 pandemic (March 2020).

Intraday arbitrage portfolio robustness (Internet Appendix IA.6). We evaluate the performance of the intraday arbitrage portfolio along six dimensions.

(I) **FACTOR CHOICE:** Annual Sharpe ratios range from 1.99 to 2.61 across a broad grid of (K_C, K_J) specifications with $K_C > 0$. Portfolios with $K_J = 0$ achieve slightly higher Sharpe ratios, but risk-adjusted alphas are nearly identical once standard factor models are applied. This suggests that jump decomposition separates risk components rather than improving ex-post performance.

(II) **INCREMENTAL VALUE OVER KIM ET AL. (2021):** Our portfolio retains highly significant alphas after controlling for their low-frequency PPCA portfolio, indicating that explicit high-frequency jump decomposition adds information beyond what low-frequency approaches capture.

(III) **ESTIMATION WINDOW:** Sharpe ratios decline monotonically from 2.49 at a one-month window to 0.33 at twelve months, consistent with short-lived mispricing signals and highlighting the value of short estimation horizons.

(IV) **REBALANCING FREQUENCY:** Sharpe ratios of 1.15–1.49 under daily rebalancing and 1.11–1.44 under monthly rebalancing show that the signals persist under realistic trading constraints.

(V) **LIQUIDITY:** Applying stricter liquidity screens yields Sharpe ratios of 1.79–2.26, indicating that the results are not driven by illiquid stocks.

(VI) **SAMPLING FREQUENCY AND JUMP DETECTION:** Results remain robust when using 15- and 60-minute intervals and when applying the Lee and Wang (2020) intraday volatility adjustment.

Robustness of risk premia estimation results (Internet Appendix IA.7). We conduct two additional robustness tests.

(I) **ALTERNATIVE FACTOR SPECIFICATIONS:** Diffusive risk premia and the asymmetric pricing of signed jump risk remain robust across alternative factor dimensions supported by

the PER test, including $(K_C = 4, K_J = 1)$ and $(K_C = 5, K_J = 2)$. We also report pricing results using unanchored statistical factors.

(II) JUMP DETECTION ROBUST TO INTRADAY VOLATILITY PATTERNS: Applying the Lee and Wang (2020) volatility-pattern-adjusted jump detection yields similar pricing results, confirming that our conclusions are not driven by spurious jump detections.

5 Conclusion

This paper provides a framework that integrates firm characteristics with high-frequency return panels to decompose systematic risk into diffusive and jump components and identify mispricing. Extending Projected PCA to decomposed intraday diffusive and jump return panels yields characteristic-based risk exposures for each component and an associated mispricing measure. The framework accommodates a broad class of continuous-time multifactor models and provides consistent estimators for decomposed risk exposures, mispricing, and the resulting intraday arbitrage portfolio.

Applied to more than 6,600 U.S. stocks from 2000–2021, the framework delivers four main findings. First, systematic diffusive and jump risks exhibit markedly different dynamics: diffusive risk is pervasive and relatively stable, whereas systematic jump risk is episodic and spikes during major market disruptions. Second, linking decomposed risks and mispricing to the characteristic clusters of Jensen et al. (2023) reveals a clear hierarchy. Systematic diffusive risk concentrates in classical risk attributes, systematic jump risk draws on a broader and more time-varying set of firm fundamentals, and mispricing is the most dispersed across characteristics, highlighting distinct economic forces behind each component. Third, an intraday arbitrage portfolio based on estimated mispricing delivers an annualized Sharpe ratio of 2.49 with monthly alphas unexplained by standard factor models and remains resilient during the global financial crisis and the COVID-19 pandemic. Fourth, both diffusive and jump risks command significant premia, with pronounced asymmetry in jump risk compen-

sation. These premia can be traced to their characteristic sources, with jump risk premia dispersed across several fundamentals-related clusters.

More broadly, the results highlight the value of combining firm characteristics with high-frequency return data to study the structure of systematic risk and mispricing. By decomposing systematic exposures into diffusive and jump components within a characteristic-based framework, the approach provides a new lens for understanding how different economic forces shape expected returns. Future research may apply the framework to other asset classes, such as derivatives, or examine the joint dynamics across multiple markets, including their cross-autocorrelation structures. Integrating high-frequency characteristic-based estimation with structural asset pricing models may further deepen our understanding of the economic mechanisms linking firm characteristics, systematic risks, and expected returns in modern financial markets.

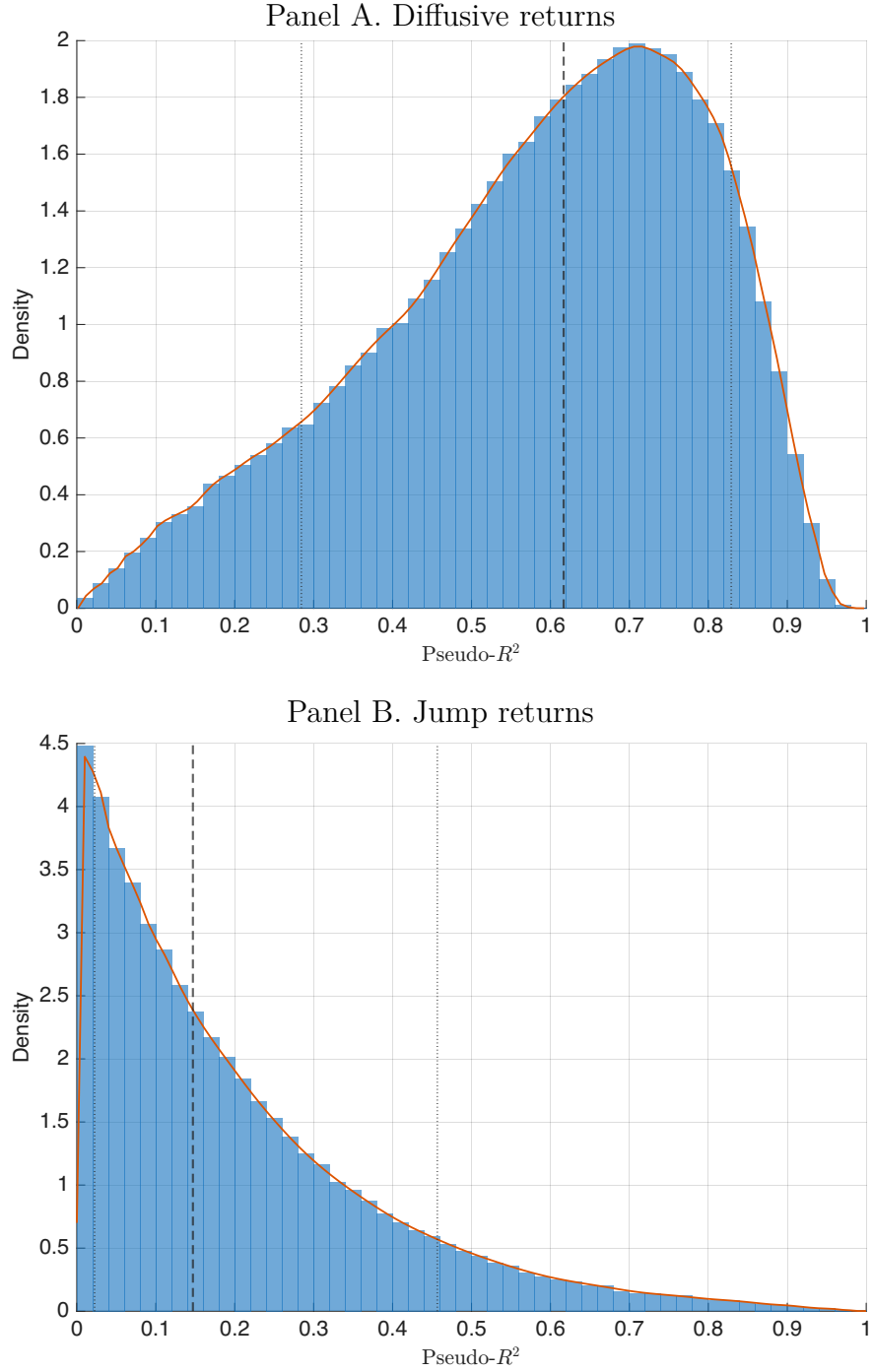
References

- Aït-Sahalia, Yacine, Jean Jacod, and Dacheng Xiu, 2025, Continuous-time fama-macbeth regressions, *The Review of Financial Studies* 38, 3542–3579.
- Aït-Sahalia, Yacine, and Dacheng Xiu, 2019, Principal component analysis of high-frequency data, *Journal of the American Statistical Association* 114, 287–303.
- Aleti, Saketh, and Tim Bollerslev, 2025, News and asset pricing: A high-frequency anatomy of the sdf, *The Review of Financial Studies* 38, 712–759.
- Andersen, Torben G., and Tim Bollerslev, 1997, Intraday periodicity and volatility persistence in financial markets, *Journal of Empirical Finance* 4, 115–158.
- Bakshi, Gurdip, Charles Cao, and Zhiwu Chen, 1997, Empirical Performance of Alternative Option Pricing Models, *Journal of Finance* 52, 2003–2049.
- Bandi, Federico M., and Roberto Renò, 2022, β in the tails, *Journal of Econometrics* 227, 134–150.
- Barberis, Nicholas, and Ming Huang, 2008, Stocks as lotteries: The implications of probability weighting for security prices, *American Economic Review* 98, 2066–2100.
- Barro, Robert J., 2006, Rare disasters and asset markets in the twentieth century, *The Quarterly Journal of Economics* 121, 823–866.
- Bollerslev, Tim, Sophia Zhengzi Li, and Viktor Todorov, 2016, Roughing up beta: Continuous versus discontinuous betas and the cross section of expected stock returns, *Journal of Financial Economics* 120, 464–490.
- Bollerslev, Tim, and Viktor Todorov, 2011, Tails, fears, and risk premia, *Journal of Finance* 66, 2165–2211.
- Bollerslev, Tim, Viktor Todorov, and Lai Xu, 2015, Tail risk premia and return predictability, *Journal of Financial Economics* 118, 113–134.
- Carhart, Mark M, 1997, On persistence in mutual fund performance, *Journal of Finance* 52, 57–82.
- Chamberlain, Gary, and Michael Rothschild, 1983, Arbitrage, factor structure, and mean-variance analysis on large asset markets, *Econometrica* 1281–1304.
- Chen, Dachuan, Per A. Mykland, and Lan Zhang, 2020, The five trolls under the bridge: Principal component analysis with asynchronous and noisy high frequency data, *Journal of the American Statistical Association* 115, 1960–1977.
- Chordia, Tarun, Amit Goyal, and Jay A. Shanken, 2017, Cross-sectional asset pricing with individual stocks: Betas versus characteristics, Technical report, Emory University.

- Connor, Gregory, Matthias Hagmann, and Oliver Linton, 2012, Efficient semiparametric estimation of the Fama–French model and extensions, *Econometrica* 80, 713–754.
- Connor, Gregory, and Robert A Korajczyk, 1993, A test for the number of factors in an approximate factor model, *Journal of Finance* 48, 1263–1291.
- Daniel, Kent, and Sheridan Titman, 1997, Evidence on the characteristics of cross-sectional variation in stock returns, *Journal of Finance* 52, 1–33.
- Duffie, Darrell, Jun Pan, and Kenneth Singleton, 2000, Transform analysis and asset pricing for affine jump-diffusions, *Econometrica* 68, 1343–1376.
- Eraker, Bjørn, Michael Johannes, and Nicholas Polson, 2003, The Impact of Jumps in Volatility and Returns, *Journal of Finance* 58, 1269–1300.
- Fama, Eugene F, and Kenneth R French, 1993, Common risk factors in the returns on stocks and bonds, *Journal of Financial Economics* 33, 3–56.
- Fama, Eugene F, and Kenneth R French, 2015, A five-factor asset pricing model, *Journal of Financial Economics* 116, 1–22.
- Fama, Eugene F., and Kenneth R. French, 2018, Choosing factors, *Journal of Financial Economics* 128, 234–252.
- Fan, J., Y. Liao, and W Wang, 2016, Projected principal component analysis in factor models, *Annals of Statistics* 44, 219–254.
- Ferson, Wayne E, and Campbell R Harvey, 1997, Fundamental determinants of national equity market returns: A perspective on conditional asset pricing, *Journal of Banking & Finance* 21, 1625 – 1665.
- Frazzini, Andrea, and Lasse Heje Pedersen, 2014, Betting against beta, *Journal of Financial Economics* 111, 1–25.
- Jacod, Jean, Huidi Lin, and Viktor Todorov, 2022, Systematic jump risk, Working paper, Northwestern University.
- Jensen, Theis Ingerslev, Bryan T. Kelly, and Lasse Heje Pedersen, 2023, Is there a replication crisis in finance?, *Journal of Finance* 78, 2465–2518.
- Kelly, Bryan, Seth Pruitt, and Yinan Su, 2019, Characteristics are covariances: A unified model of risk and return, *Journal of Financial Economics* 134, 501–524.
- Kim, Soohun, Robert A Korajczyk, and Andreas Neuhierl, 2021, Arbitrage portfolios, *Review of Financial Studies* 34, 2813–2856.
- Kim, Soohun, and Georgios Skoulakis, 2018, Ex-post risk premia estimation and asset pricing tests using large cross sections: The regression-calibration approach, *Journal of Econometrics* 204, 159–188.

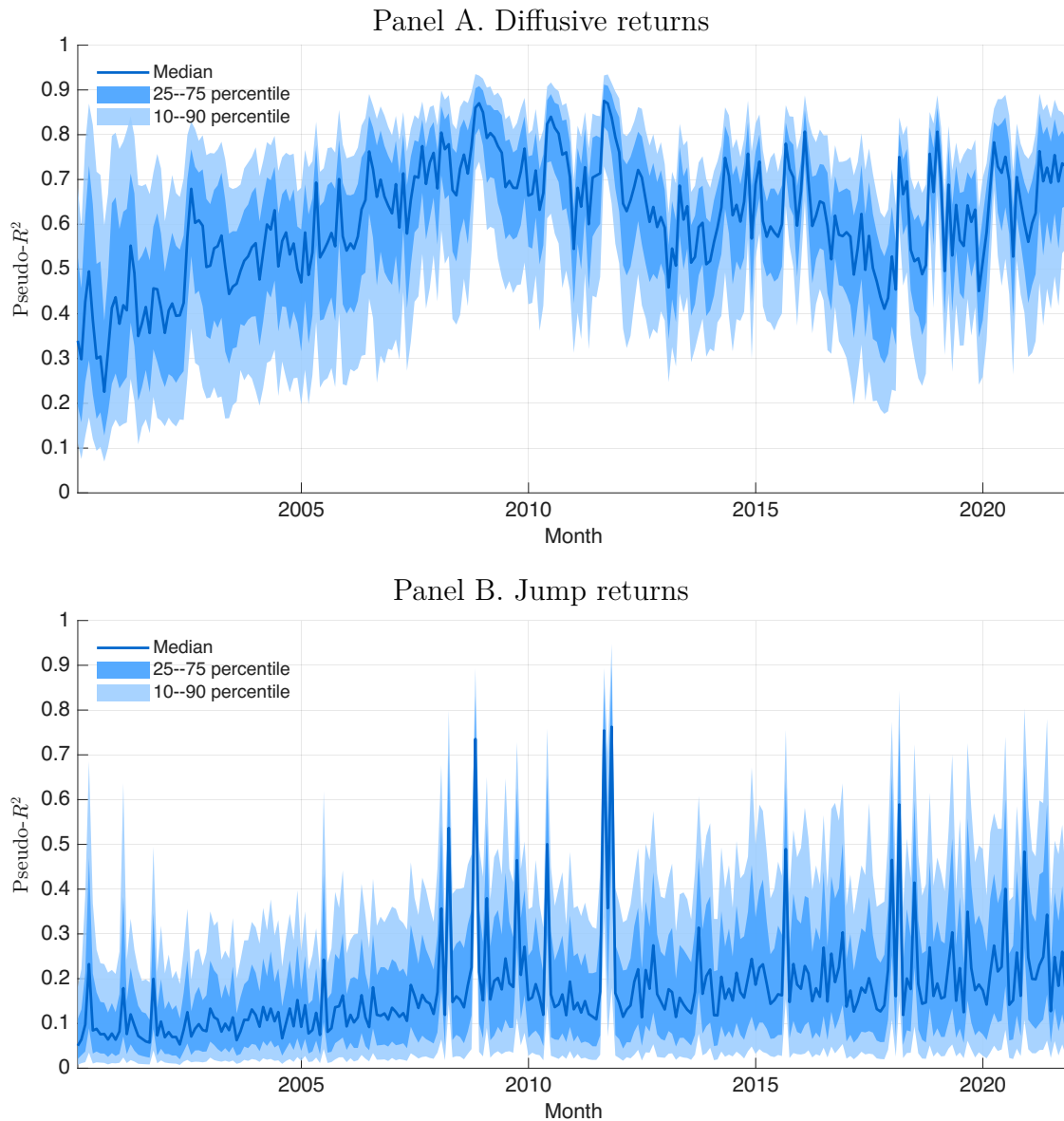
- Kozak, Serhiy, Stefan Nagel, and Shrihari Santosh, 2018, Interpreting factor models, *Journal of Finance* 73, 1183–1223.
- Lee, Suzanne S., 2012, Jumps and Information Flow in Financial Markets, *Review of Financial Studies* 25, 439–479.
- Lee, Suzanne S, and Per A Mykland, 2008, Jumps in financial markets: A new nonparametric test and jump dynamics, *Review of Financial Studies* 21, 2535–2563.
- Lee, Suzanne S., and Minho Wang, 2019, The impact of jumps on carry trade returns, *Journal of Financial Economics* 131, 433–455.
- Lee, Suzanne S., and Minho Wang, 2020, Tales of tails: Jumps in currency markets, *Journal of Financial Markets* 48, None.
- Li, Jia, Viktor Todorov, and George Tauchen, 2019, Jump factor models in large cross-sections, *Quantitative Economics* 10, 419–456.
- Liu, Weimin, 2006, A liquidity-augmented capital asset pricing model, *Journal of Financial Economics* 82, 631–671.
- Merton, Robert C., 1973, An intertemporal capital asset pricing model, *Econometrica* 41, 867–887.
- Murtagh, Fionn, and Pierre Legendre, 2014, Ward’s hierarchical agglomerative clustering method: Which algorithms implement Ward’s criterion?, *Journal of Classification* 31, 274–295.
- Pan, Jun, 2002, The jump-risk premia implicit in options: evidence from an integrated time-series study, *Journal of Financial Economics* 63, 3–50.
- Pelger, Markus, 2019, Large-dimensional factor modeling based on high-frequency observations, *Journal of Econometrics* 208, 23–42.
- Pelger, Markus, 2020, Understanding Systematic Risk: A High-Frequency Approach, *Journal of Finance* 75, 2179–2220.
- Raponi, Valentina, Cesare Robotti, and Paolo Zaffaroni, 2020, Testing beta-pricing models using large cross-sections, *Review of Financial Studies* 33, 2796–2842.
- Ross, Stephen A, 1976, The arbitrage theory of capital asset pricing, *Journal of Economic Theory* 13, 341–360.
- Ward, Joe H., 1963, Hierarchical grouping to optimize an objective function, *Journal of the American Statistical Association* 58, 236–244.
- Yu, Seunghyeon, Viktor Todorov, and Torben G. Andersen, 2024, The factor structure of systematic jump risk, Working paper, Northwestern University.

Figure 1: Pooled cross-sectional distribution of the pseudo- R^2 for diffusive and jump returns



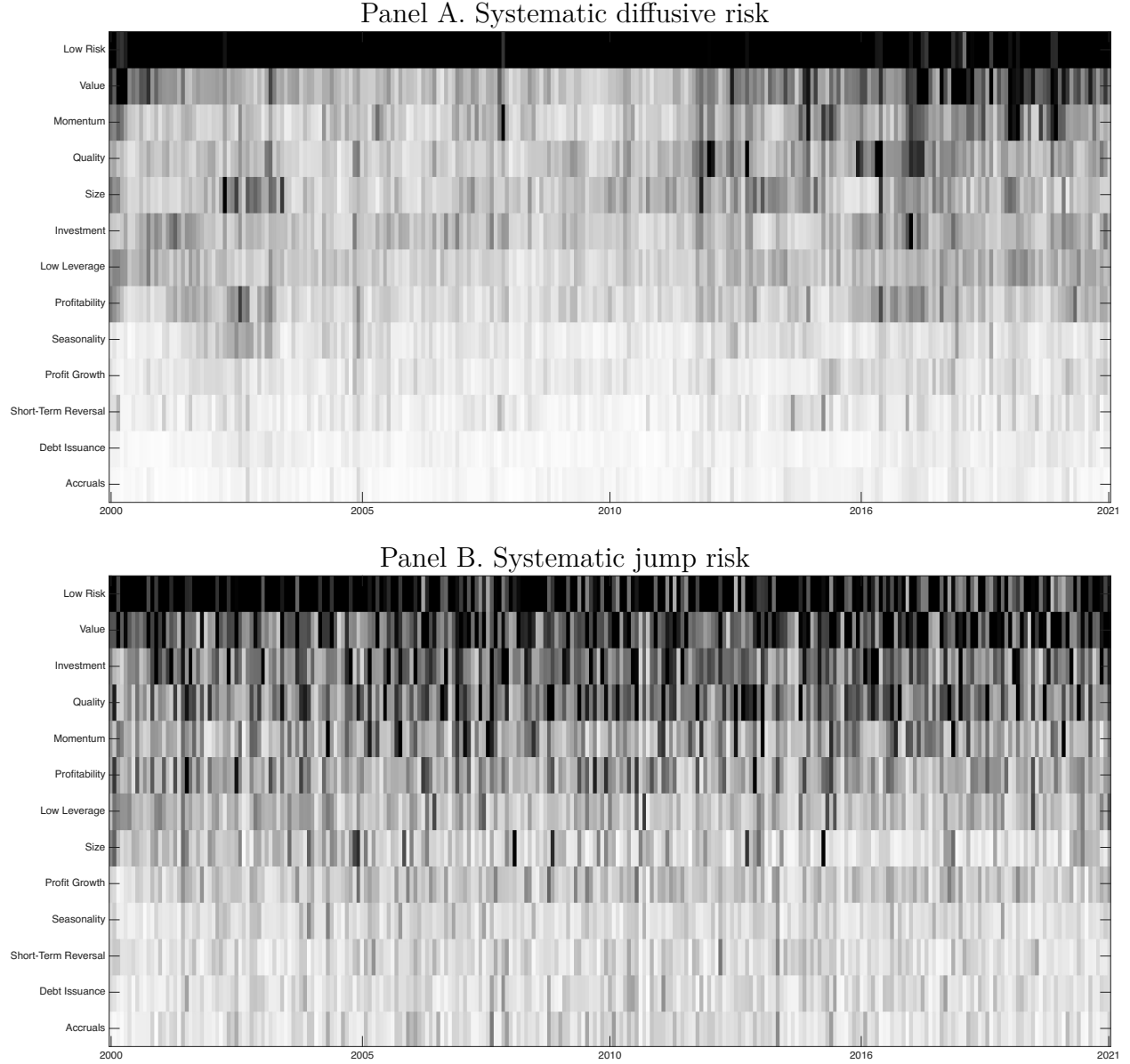
This figure plots the pooled cross-sectional distributions of the pseudo- R^2 computed as the fraction of the total realized variance of projected diffusive and jump returns explained by the systematic components. Panel A and Panel B plot the distributions of the pseudo- R^2 for the systematic diffusive component and the systematic jump component, respectively. Each histogram is normalized to integrate to one, and the smooth curve is a kernel density estimate. Vertical dashed lines indicate the median and the 10th and 90th percentiles.

Figure 2: Time-series of cross-sectional distribution of pseudo- R^2 for diffusive and jump returns



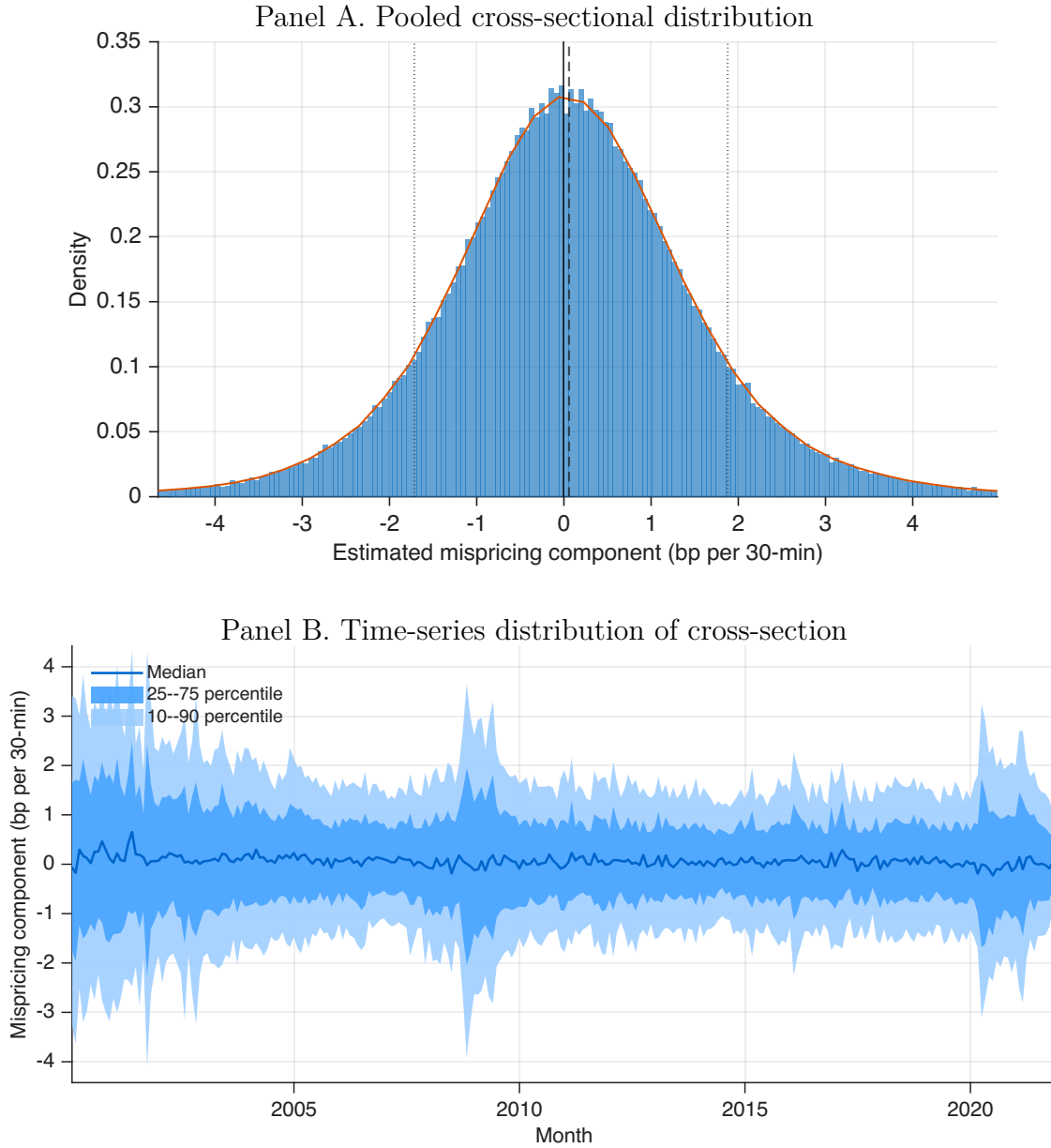
This figure plots the time-series of the cross-sectional distribution of the pseudo- R^2 computed as the fraction of the total realized variance of projected diffusive and jump returns explained by the systematic components. Panel A and Panel B plot the time-series of the distribution of pseudo- R^2 s for the systematic diffusive component and the systematic jump component, respectively. Solid line represents the median in each month, along with shaded bands for the 25–75 and 10–90 percentile ranges.

Figure 3: Cluster-level attribution of systematic diffusive and jump risks



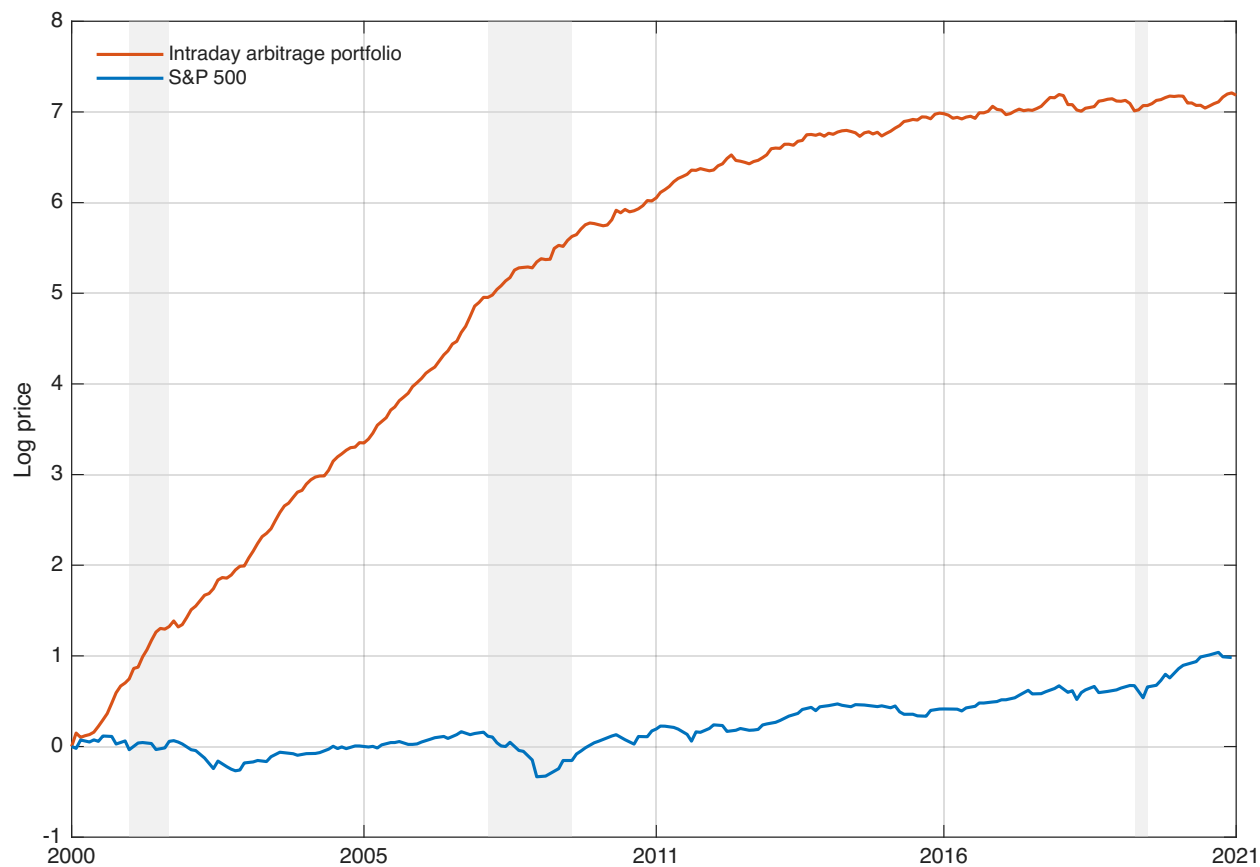
This figure displays two heat maps of cluster-level attributions of systematic diffusive risk (Panel A) and jump risk (Panel B) over time. Each row corresponds to an economic cluster defined by the taxonomy of Jensen et al. (2023). At each month-end, cluster-level attributions are computed by aggregating normalized (absolute) characteristic attributions within each cluster, so that cluster shares sum to one across clusters. For visualization, cluster attributions are scaled within each month by the maximum cluster attribution in that month; darker shading therefore indicates greater relative importance at a given point in time. Clusters are ordered by their average attribution to diffusive risk (Panel A) and jump risk (Panel B) over the sample period.

Figure 4: Pooled cross-sectional and time-series distributions of estimated mispricing component



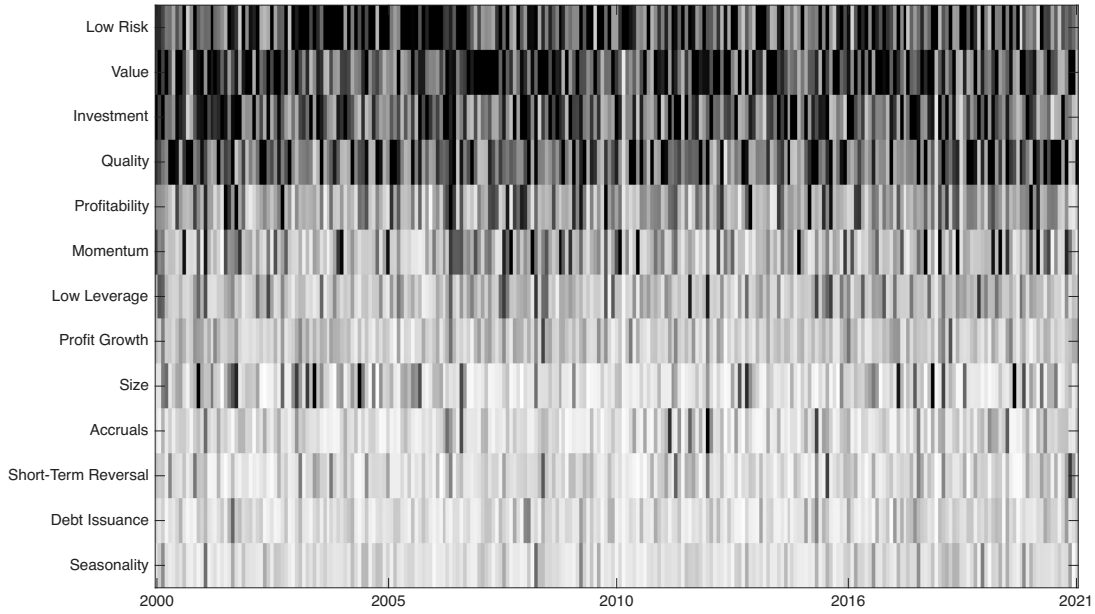
This figure summarizes the empirical properties of the estimated mispricing component. Panel A plots the pooled cross-sectional distribution of $\hat{\alpha}_i$ across all stock-months, where the histogram is normalized to integrate to one and the smooth curve is a kernel density estimate. Vertical dashed lines indicate the median and the 10th and 90th percentiles. Panel B summarizes the time-series evolution of the cross-sectional distribution of the mispricing component by plotting, in each month, the median along with shaded bands for the 25–75 and 10–90 percentile ranges of $\hat{\alpha}_i$ across stocks. Mispricing is measured in basis points per 30-minute interval.

Figure 5: Log price of intraday arbitrage portfolio



This figure plots the logarithmic price paths over the sample period from January 2000 to December 2021. The orange line represents the log price path of our intraday arbitrage portfolio, assuming an initial investment of \$1 at the beginning of the period. The light blue line shows the log price path of an investment in the S&P 500 index. Shaded regions indicate NBER recession periods.

Figure 6: Cluster-level attribution of mispricing



This figure displays a heatmap of cluster-level attributions of mispricing over time. Each row corresponds to an economic cluster defined by the taxonomy of Jensen et al. (2023). At each month-end, cluster-level attributions are computed by aggregating normalized (absolute) characteristic attributions within each cluster, so that cluster shares sum to one across clusters. For visualization, cluster attributions are scaled within each month by the maximum cluster attribution in that month; darker shading therefore indicates greater relative importance at a given point in time. Clusters are ordered by their average attribution to mispricing over the sample period.

Table 1: Data and Summary Statistics

Panel A: Sample structure				
Sample period	January 2000 – December 2021			
	Mean	P25	Median	P75
# stocks per month	1,336	1,156	1,230	1,465
# HF return observations per month	364,139	313,072	340,886	402,370
Total stock-months	352,595			
Total HF return observations	96,132,699			
Panel B: Liquidity / tradability				
Price	42.60	14.33	26.96	47.40
Trading volume (monthly, in thousands)	30,658	2,979	8,359	24,494
Dollar volume (monthly, \$ mil)	1,376	59	238	948
Shares outstanding (in thousands)	164,014	22,116	46,295	113,459
Market equity ME (\$ mil)	8,232	413	1,267	4,338
Panel C: Diffusive / jump decomposition				
Realized variation (RV)	0.0191	0.0048	0.0094	0.0198
Continuous variation (CV)	0.0152	0.0041	0.0079	0.0161
Jump variation (JV)	0.0040	0.0000	0.0008	0.0030
Jump share (JV/RV)	0.1479	0.0000	0.1053	0.2240
Jump count per month	1.7910	0	1	2
Mean jump (conditional)	0.0372	0.0218	0.0312	0.0456

This table provides summary statistics for the main sample. Panel A summarizes sample coverage and the scale of the high-frequency data, Panel B reports liquidity and size characteristics, and Panel C presents the decomposition of realized return variation into diffusive and jump components.

Table 2: Intraday arbitrage portfolio returns and common risk factors

Panel A. Full sample (Jan 2000 to Dec 2021)					
	CAPM	FF3	FF3+UMD	FF5	FF5+UMD
intercept	0.0292*** (11.92)	0.0288*** (12.00)	0.0283*** (11.82)	0.0277*** (10.97)	0.0275*** (10.94)
MKT	-0.1173** (-2.18)	-0.1750*** (-3.19)	-0.1232** (-2.06)	-0.1296** (-2.14)	-0.0926 (-1.46)
SMB		0.2952*** (3.71)	0.2741*** (3.45)	0.3009*** (3.29)	0.2758*** (3.00)
HML		-0.0145 (-0.20)	0.0257 (0.35)	-0.1424 (-1.47)	-0.0862 (-0.85)
UMD			0.1078** (2.14)		0.0940* (1.84)
RMW				0.0580 (0.53)	0.0361 (0.33)
CMA				0.3088** (2.03)	0.2701* (1.77)
Adj. R2 (%)	1.4091	5.6741	6.9605	6.5294	7.3921
Obs	263	263	263	263	263
Panel B. Post-2008 sample (Jan 2009 to Dec 2021)					
intercept	0.0126*** (4.65)	0.0120*** (4.39)	0.0120*** (4.41)	0.0119*** (4.26)	0.0119*** (4.29)
MKT	-0.0328 (-0.55)	-0.0315 (-0.48)	-0.0111 (-0.16)	-0.0383 (-0.56)	-0.0191 (-0.28)
SMB		0.1768 (1.61)	0.1885* (1.71)	0.2029* (1.70)	0.2209* (1.85)
HML		-0.2040** (-2.27)	-0.1614* (-1.69)	-0.1884* (-1.74)	-0.1310 (-1.13)
UMD			0.0824 (1.28)		0.0907 (1.39)
RMW				0.0890 (0.59)	0.1060 (0.70)
CMA				-0.0710 (-0.37)	-0.1093 (-0.56)
Adj. R2 (%)	-0.4541	2.1821	2.5944	1.1778	1.7835
Obs	156	156	156	156	156

The table presents time series regression results of the intraday arbitrage portfolio returns on common risk factors. It reports the intercept (which corresponds to a monthly alpha) and factor loadings on well-known risk factors from Fama and French (1993), Carhart (1997), and Fama and French (2015). t -statistics are provided in parentheses. Panel A covers the sample period from January 2000 to December 2021, while Panel B covers January 2009 to December 2021. Significance levels are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

Table 3: Pricing of systematic diffusive and jump factors

	(1)		(2)	
	Risk premium	(<i>t</i> -stat)	Risk premium	(<i>t</i> -stat)
<i>Diffusive factors</i>				
$\mathbf{f}_C^{\text{MKT}}$	0.6196***	(2.631)	0.5253**	(2.312)
$\mathbf{f}_C^{\text{SMB}}$	0.5393***	(4.478)	0.6038***	(5.078)
$\mathbf{f}_C^{\text{HML}}$	0.3543***	(3.416)	0.4128***	(4.054)
$\mathbf{f}_C^{\text{Orth}}$	0.4911	(1.100)	0.4809	(1.078)
<i>Jump factors</i>				
$\mathbf{f}_J^{\text{MKT}}$	0.7523**	(2.474)		
$\mathbf{f}_J^{\text{MKT_Pos}}$			−0.2993	(−1.401)
$\mathbf{f}_J^{\text{MKT_Neg}}$			1.1803***	(4.910)
$\mathbf{f}_J^{\text{Orth}}$	0.5518*	(1.734)		
$\mathbf{f}_J^{\text{Orth_Pos}}$			−0.4955**	(−2.338)
$\mathbf{f}_J^{\text{Orth_Neg}}$			1.2004***	(4.020)
R^2 (%)	7.560		8.290	

This table reports pricing results for anchored systematic diffusive and jump factors constructed as described in Section 4.2. The results are obtained using the continuous-time Fama–MacBeth (1973) regression framework of Aït-Sahalia et al. (2025). The first two columns under (1) report risk premia for three diffusive factors corresponding to the market, size, and value factors ($\mathbf{f}_C^{\text{MKT}}$, $\mathbf{f}_C^{\text{SMB}}$, and $\mathbf{f}_C^{\text{HML}}$), one orthogonal diffusive factor ($\mathbf{f}_C^{\text{Orth}}$), and two jump factor capturing market-wide and orthogonal jump risk ($\mathbf{f}_J^{\text{MKT}}$ and $\mathbf{f}_J^{\text{Orth}}$). The last two columns under (2) report results from an extended specification in which jump factors are further decomposed into positive and negative jump components. Risk premium estimates are multiplied by 100. *t*-statistics, computed using the continuous-time variance estimator of Aït-Sahalia et al. (2025), are shown in parentheses. The last row (R^2) reports the time-series average of the cross-sectional regression R^2 s. Significance levels are indicated by ***, **, and * for 1%, 5%, and 10%, respectively.

Table 4: Cluster-level attribution of systematic diffusive risk premia

Cluster	Mkt ^C			Smb ^C			Hml ^C			Orth ^C		
	$\hat{\lambda}$ (bps)	Share (%)	<i>t</i> -stat	$\hat{\lambda}$ (bps)	Share (%)	<i>t</i> -stat	$\hat{\lambda}$ (bps)	Share (%)	<i>t</i> -stat	$\hat{\lambda}$ (bps)	Share (%)	<i>t</i> -stat
Low Risk	28.95	55.1	2.33**	11.44	19.0	4.70***	8.19	19.8	3.95***	7.95	16.5	1.15
Size	1.52	2.9	0.65	17.10	28.3	5.21***	5.45	13.2	3.27***	0.35	0.7	0.07
Low Leverage	3.74	7.1	1.97**	3.83	6.3	3.97***	3.94	9.5	3.34***	11.01	22.9	1.32
Quality	-1.32	-2.5	-1.01	5.79	9.6	3.06***	4.63	11.2	3.29***	9.57	19.9	1.25
Momentum	5.25	10.0	2.70***	4.93	8.2	4.37***	5.18	12.6	3.88***	3.79	7.9	0.65
Profit Growth	0.62	1.2	0.81	2.28	3.8	3.52***	2.70	6.5	3.05***	7.91	16.4	1.79*
Value	6.67	12.7	3.17***	2.19	3.6	1.30	1.14	2.8	0.87	-0.82	-1.7	-0.14
Investment	1.05	2.0	0.97	3.63	6.0	4.10***	2.92	7.1	3.17***	2.30	4.8	0.43
Seasonality	1.47	2.8	0.66	3.59	5.9	3.99***	2.27	5.5	1.74*	1.44	3.0	0.63
Profitability	2.68	5.1	1.16	2.18	3.6	2.68***	1.52	3.7	1.98**	1.59	3.3	0.52
Short-Term Reversal	1.71	3.3	1.77*	1.78	3.0	3.67***	1.92	4.6	2.90***	1.07	2.2	0.33
Debt Issuance	0.54	1.0	1.18	0.67	1.1	2.42**	0.78	1.9	2.06**	1.03	2.1	0.88
Accruals	-0.36	-0.7	-0.87	0.96	1.6	2.19**	0.63	1.5	2.15**	0.89	1.9	0.68
Total	52.53	100.0	2.31**	60.38	100.0	5.08***	41.28	100.0	4.05***	48.09	100.0	1.08

For each diffusive factor, $\hat{\lambda}$ denotes the risk premium (in basis points per month) attributable to each characteristic cluster, and Share is the cluster's percentage contribution to the total risk premium. *t*-statistics are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 5: Clueter-level attribution of systematic jump risk premia

Cluster	Mkt ^{J_Pos}				Mkt ^{J_Neg}				Orth ^{J_Pos}				Orth ^{J_Neg}			
	$\hat{\lambda}$ (bps)	Share (%)	t -stat		$\hat{\lambda}$ (bps)	Share (%)	t -stat		$\hat{\lambda}$ (bps)	Share (%)	t -stat		$\hat{\lambda}$ (bps)	Share (%)	t -stat	
Low Risk	-14.52	48.5	-1.36		45.07	38.2	3.83***		-6.95	14.0	-1.53		16.99	14.2	3.21***	
Value	-0.65	2.2	-0.27		11.12	9.4	3.21***		-8.93	18.0	-1.60		20.82	17.3	3.18***	
Low Leverage	-1.43	4.8	-0.66		13.18	11.2	4.27***		-7.75	15.6	-1.96**		13.64	11.4	3.51***	
Momentum	-1.15	3.9	-0.81		3.94	3.3	2.72***		-5.41	10.9	-2.41**		16.35	13.6	2.20**	
Size	-1.25	4.2	-0.26		19.22	16.3	7.02***		-1.18	2.4	-0.99		3.54	3.0	1.46	
Quality	-1.02	3.4	-0.46		3.62	3.1	1.82*		-3.60	7.3	-1.95*		16.64	13.9	3.65***	
Investment	-3.34	11.2	-1.82*		2.75	2.3	1.45		-6.05	12.2	-2.47**		10.02	8.3	2.96***	
Seasonality	-1.06	3.5	-0.68		3.59	3.0	3.46***		-4.03	8.1	-2.37**		11.24	9.4	3.21***	
Profitability	-3.34	11.1	-1.86*		10.64	9.0	4.07***		-0.70	1.4	-0.29		1.96	1.6	0.92	
Profit Growth	-0.95	3.2	-1.07		1.38	1.2	1.37		-1.76	3.6	-1.43		4.82	4.0	3.19***	
Short-Term Reversal	-0.60	2.0	-0.89		1.34	1.1	1.88*		-1.30	2.6	-1.57		4.85	4.0	2.82***	
Debt Issuance	-0.01	0.0	-0.01		1.39	1.2	1.86*		-1.27	2.6	-0.94		-2.45	-2.0	-2.06**	
Accruals	-0.61	2.0	-0.88		0.77	0.7	1.37		-0.62	1.3	-0.76		1.61	1.3	2.13**	
Total	-29.93	100.0	-1.40		118.03	100.0	4.91***		-49.55	100.0	-2.34**		120.04	100.0	4.02***	

For each jump factor, $\hat{\lambda}$ denotes the risk premium (in basis points per month) attributable to each characteristic cluster, and Share is the cluster's percentage contribution to the total risk premium. t -statistics are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.